

M. Tech Dissertation

On

**Investigating Agriculture Demand Side Management through ESCO
– (A case study of Solapur Pump replacement Project)**

Submitted in partial fulfillment for the requirement of Degree of

M. Tech. in Technology and Development

by

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Under the Supervision of

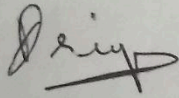
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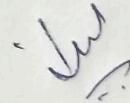
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APPROVAL SHEET

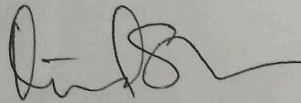
This is to certify that the project entitled "Investigating Agriculture Demand Side Management through ESCO – (A case study of Solapur Pump replacement Project)" prepared by Saurabh Khobaragade is approved for degree of Master of Technology at Centre for Technology Alternatives for Rural Areas (CTARA), IIT Bombay.



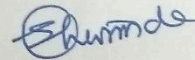
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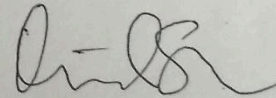
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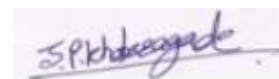
DECLARATION

I hereby declare that the report entitled **“Investigating Agriculture Demand Side Management through ESCO – (A case study of Solapur Pump replacement Project).”** submitted by me, for the partial fulfillment of the degree of Master of Technology to CTARA, IITB is a record of the M.Tech Project work carried out by me under the supervision of Dr. Priya Jadhav, Assistant Professor, CTARA and Dr. Vishal Sardeshpande, Adjunct professor CTARA.

I further declare that this written submission represents my ideas in my own words and where other's ideas or words have been included, I have adequately cited and referenced the original sources. I affirm that I have adhered to all principles of academic honesty and integrity and have not misrepresented or falsified any idea/data/fact/source to the best of my knowledge. I understand that any violation of the above will cause for disciplinary action by the Institute and can also evoke penal action from the sources which have not been cited properly.

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A handwritten signature in blue ink, appearing to read 'S.P. Khedekar', is written over a light purple rectangular background.

Signature of the Candidate

ACKNOWLEDGEMENT

I am highly indebted to my project mentor, Prof. Dr. Priya Jadhav, for her continuous supervision, motivation and guidance throughout the tenure of this project in spite of her hectic schedule. She has truly remained my driving spirit in the project and her experience gave me a light in handling the intricacies of the project and helped me in identifying abstruse concepts requiring knowledge and perception, handling critical situations and in understanding the objective of work.

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ABSTRACT

Agriculture is one of the largest electricity consuming sector in India. Studies have shown that less than 2% of irrigation pumpsets have efficiency greater than 40%. Realizing this huge energy saving potential in the sector, various projects have been implemented under AgDSM (Agriculture Demand Side Management) in the last four decades. Pump replacement, is one of such initiative expanding rapidly in recent timeline. This initiative involves replacing old inefficient pumps with BEE (Bureau of Energy Efficiency) star rated energy efficient ones with free maintenance for specified project duration; with the help of ESCOs (Energy Service Companies).

The government implemented a pilot pump replacement and maintenance project in Solapur from 2010 to 2017. Based on the outcomes of this project, similar initiatives are being taken across different states. This is now in expansion stage with a big proposal to replace 2lakh IPS (Irrigation Pumpset) across India. Though the Solapur project is claimed to be a success, there is no formal document publicly available yet. This report focuses on analysing this project; to find the scope of improvement. It includes assessing the pilot project on technical, financial and social front in brief.

Actual performance of IPS in the field is very different from the designed or expected one due to varying operating conditions. The information on effect of external factors on performance and usage of pumpsets in field is limited. This limitation usually arises from the variation in external factors itself with locations, pump usage pattern, availability of real data etc. We tried to study the effect of these factors with pilot project as best as is possible with the data available.

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ABBREVIATIONS

AgDSM	Agriculture Demand Side Management
BEP	Best Efficiency Point
DISCOM	Distribution Company
DPR	Detailed Project Report
EEPS	Energy efficient PumpSet
ESCO	Energy Service company
GDP	Gross Domestic Product
H&Q	Head and Discharge
HP	Horse Power
IPS	Irrigation PumpSet
M&V	Monitoring and Verification
mbgl	Meters below ground level
MSEDCL	Maharashtra State Electricity Distribution Company Limited

MU	Million Units
VSD	Variable Speed Drive
Pilot Project Scheme	Used to refer the Pilot pump replacement Project in Solapur

Chapter 1 INTRODUCTION

1.1 Background:

India ranks second most in terms of agricultural production. Agriculture accounts for about 60.5% of total land i.e., 1799 lakh hectares. Farming is the biggest industry in India and plays a key role in the socio-economic development of the country. (CGIAR 2017) It employs about 50% of workforce in the country. About 20.6 million electric IPS has been energized in India, which accounts for 17.3% electricity consumption in year 2015-16. (MOSPI, 2017) The burden of subsidy on power by agriculture is increasing steadily and was about Rs 67,000 crore in year 2013-14. (Ghosh, A., & Agrawal, S. 2015) Total energy consumption in agriculture sector is estimated to show a growth of 2.4% per year till year 2020, whereas the consumption of water for irrigation is to grow by 22% by year 2025. (WENEXA 2007) This mean extensive increase in energy demand. Low tariff and flat rate charges to farmers account for an estimated 10% of the total cost of supply, about INR 240 billion a year. (World Bank 2005) This contributes to about 6% of country's fiscal deficit in year 2010-11.

Generally, the shortage conditions prevailed in the India both in terms of energy and peaking availability as 2.1% & 3.2% during year 2014-15 respectively. (CEA 2017) The study also found that only 2% of the pumps surveyed had efficiency levels above 40%. (Phadke A., 2005) In such a situation, energy efficiency and DSM activities are considered as economical and effective tools to mitigate these issues.

Under AgDSM scheme various project has been implemented to save the energy. But many of them could not go ahead of pilot stage. Seeing at the huge potential of energy saving with energy efficient pumps, a pump replacement pilot project was launched in year 2010 in Solapur district of Maharashtra. Assuming incremental cost of efficient and higher sized pumps Rs.2000 per pump system, total investment incremental cost for 250000 efficient pumps would be Rs.500 million. But total power saving 61MW, implying an avoided investment of Rs.3.05 billion. The cost-benefit ratio is below 1:6 in this case. (Dixit, G., 1996)

Issue between Farmer – utility is one of the major cause of inefficient pumping in agricultural, figure 1.3 demonstrates a vicious cycle for same. To deal with these issue government has been implementing many projects under AgDSM.

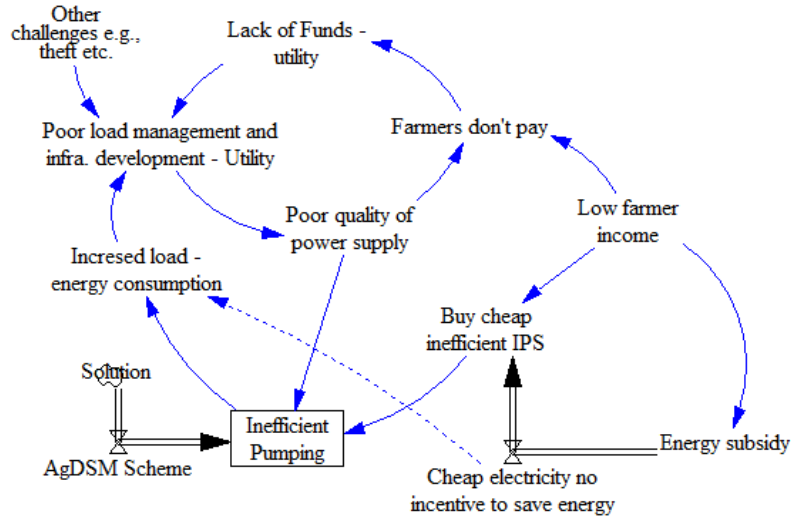


Figure 1.1 Vicious cycle of inefficient pumping

The primary focus of DSM is to save energy through activities, methodologies, and technologies that influence consumer behaviour and change their consumption patterns. But, in agriculture scenario, electricity is subsidised and available very cheaply to the farmers. As there is no incentive to farmers, they are less likely to care about energy savings. To improve the sustainability of energy savings, in Solapur pilot project 5 years of free pump maintenance is provided for replaced IPS through third party implementation.

Maharashtra is one of the progressive state in India. Majority of population is still dependent on agriculture. In addition to this Maharashtra has most number of pumpsets energized as of year 2017. About 20% of total energized pumpsets and 10% of total agriculture consumption in India, is in Maharashtra state. And about 24.5% of electricity is consumed in year 2014-15 by agriculture sector alone. MSEDCL is a state DISCOM which caters the service to about 3.17 million consumers with agricultural connected loads accounting to 19% in year 2015.

Considering all this parameters Maharashtra shows a great potential for implementation of AgDSM scheme.

From figure 1.1 & 1.2 we can see that Maharashtra, Gujarat, Karnataka & Tamilnadu alone accounts for more than 50% of pump installations across India. While Andhra Pradesh, Punjab, Rajashtan have high energy consumption in agriculture sector, with Rajashtan having about 40% of connected load from agriculture sector. Telangana, Madhya Pradesh & Andhra Pradesh acconuts for high agricultural energy consumption as well.

As Pump replacement project has concluded in year Dec. 2017, analysis of this pilot project at such an early stage (just after completion) can help us get deeper insight for similar projects in future across the country. The focus of report is to analyse the pilot project and study the field scenario of IPS performance.

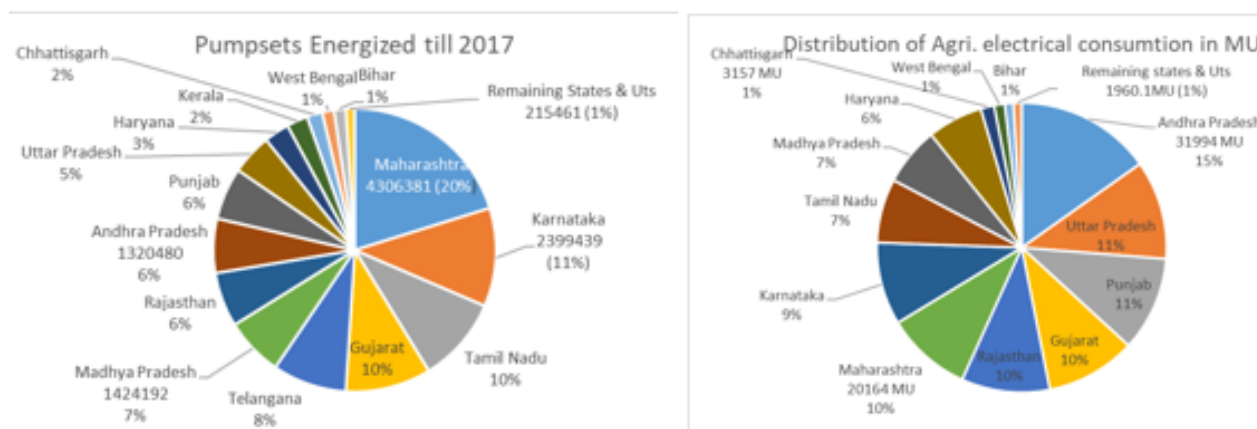


Figure 1.2 Electrified Irrigation Scenario in India

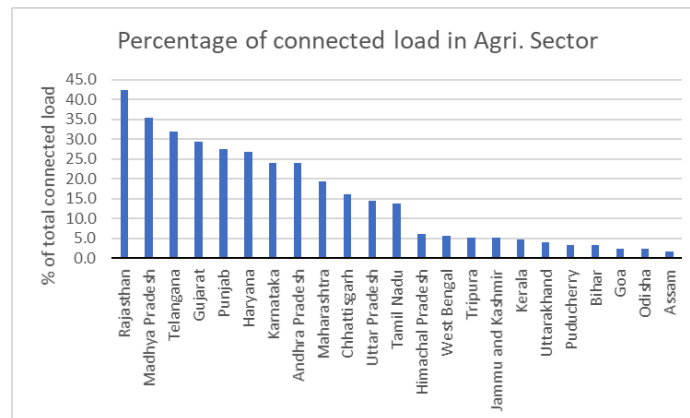


Figure 1.3 Contribution of Agricultural connected load in India

(Source: Indiatat)

1.2 Objectives

Saving the energy is basic societal concern. Looking at the huge energy saving potential in agriculture sector and need of improving the condition of energized irrigation is one way to address the issue. A pumpset replacement project is one of such step taken toward achieving the objective. Aim is to bring energy optimization in irrigation with the help of energy efficient pumpsets (EEEPs). Hence analysing this project will be worth to verify the effectiveness of this step taken, toward achieving the desired results of energy saving. In addition to this, the aim is to gain the understanding IPS usage pattern in field.

Analysing the Pump replacement project.

1. Stakeholder Analysis – Specially response of farmer to the scheme.
2. Analysing no pump related factors affecting scheme - M&V methodology & data analysis, Hours of Pumping.
3. Scheme impact analysis – To know the actual impact of the project

Studying variation in pumpset performance (efficiency).

4. Finding efficiencies of IPS in fields.
5. Studying External factors affecting their efficiencies and measurements. (Time, voltage, repair)
6. Studying performance variation of nonstandard IPS and EEPS. (e.g., variation in efficiency with voltage fluctuation.)

1.3 Structure of Report

There are many solutions available which can be implemented under AgDSM

Metering or metered tariff

Pump rectification, one-time pump replacement and Accessories replacement

Pump replacement & maintenance – ESCO involvement

This report focuses on Pump replacement through ESCOs and is mainly divided into two parts.

First part is analysis of pilot project/scheme, which includes results of field study in Mangalwedha, analysis of M&V process of scheme and Impact analysis. Field study discusses the overall preview of Irrigation scenario in Mangalwedha specifically concerning

to pilot project beneficiaries. While, M&V process itself has been analysed as this was a measure of success of project, any error in this would be significant from perspective of claimed objective of pilot project.

Second part deals with the technical aspects of irrigation pumpset (IPS), focusing on study of external factors affecting the pump performance. As, lot of data is collected for M&V process, this opportunity is used to study the pumpset performance in field. In this, the data collected is used to do as a primary analysis of effect of aging, voltage and breakdown on efficiency. Voltage is a critical parameter for farmers regarding electrical pumpset usage. Hence, a pump experiment is performed which mainly focuses variation in pumpset performance due to voltage variation; with comparative analysis of ISI and Non-ISI standardized pumpsets. Finally, the outcomes of all this analysis are compiled and discussed.

1.4 Methodology

1. Literature review:

- ☐ To get learn about overall picture of electrical pumping across India.
- ☐ Get overview of factors affecting pump performance.
- ☐ Learn about various AgDSM measures taken to improve the efficiency of IPS across country.
- ☐ To know about technological solution to improve the efficiency of IPS.
- ☐ Understanding the pilot project in depth.

2. Case Study of Solapur Pilot Project

a. Field Visits: Interaction with stakeholders, Knowing the scenario of location under study.

- To learn about aspects pertaining to usage of IPS in field.
- Survey & data collection for the investigations listed below.

b. Investigating the broader issues affecting outcomes of pilot project.

- Inappropriate pump selection is one of reason for inefficiencies – Investigating the farmers to find out priorities of their pump selection, invariably finding if this is the reason for inefficiency in the reason – Checking if awareness has increased after the implementation of pilot project.
- Checking availability of quality of power/voltage under project command area.
- Breakdown of IPS affects scheme financially, electrical faults causes major breakdown. Investigating if there is interlinkage between breakdown and Electrical panel maintenance.

c. Studying the effect of external factors on pump efficiency.

- Information about Effect of time and breakdown on efficiency is limited – Case study of selected model based on M&V data.
- Issues of Voltage is one of most relevant factor– Design of Controlled experiment.

d. Analysis of pilot project: (factors affecting the project)

1. Investigate the Hours of Pumping

- Feeder level analysis – Borale feeder analysis
- Findings from empirical study – survey of 22 beneficiaries of project.
- Metered consumer analysis – Based on Secondary data.
- Investigation based on cropping patten in region – With normal rainfall in Mangalwedha.

2. Investigate the Monitoring & verification process.

- Measurement of error in data collected by M&V.

(Analysing the 1st year and 2nd year of M&V data – Finding and quantifying the error)

- Error analysis for methodology of M&V calculation and measurements
(Comparative error analysis of primary and secondary (M&V) measurements & calculations)
- Sensitivity analysis to see how much error could have affected the measurement.
(Case study of 2 measurements taken on field)

3. Impact of the pilot project:

- Farmers response about the pilot project. – Survey of 22 farmers/ beneficiaries
- Efficiency achievements, Energy Savings. – Power measurements of Approx. 27 Farmers/beneficiaries.
- Financial aspect – Cost benefit analysis of scheme.

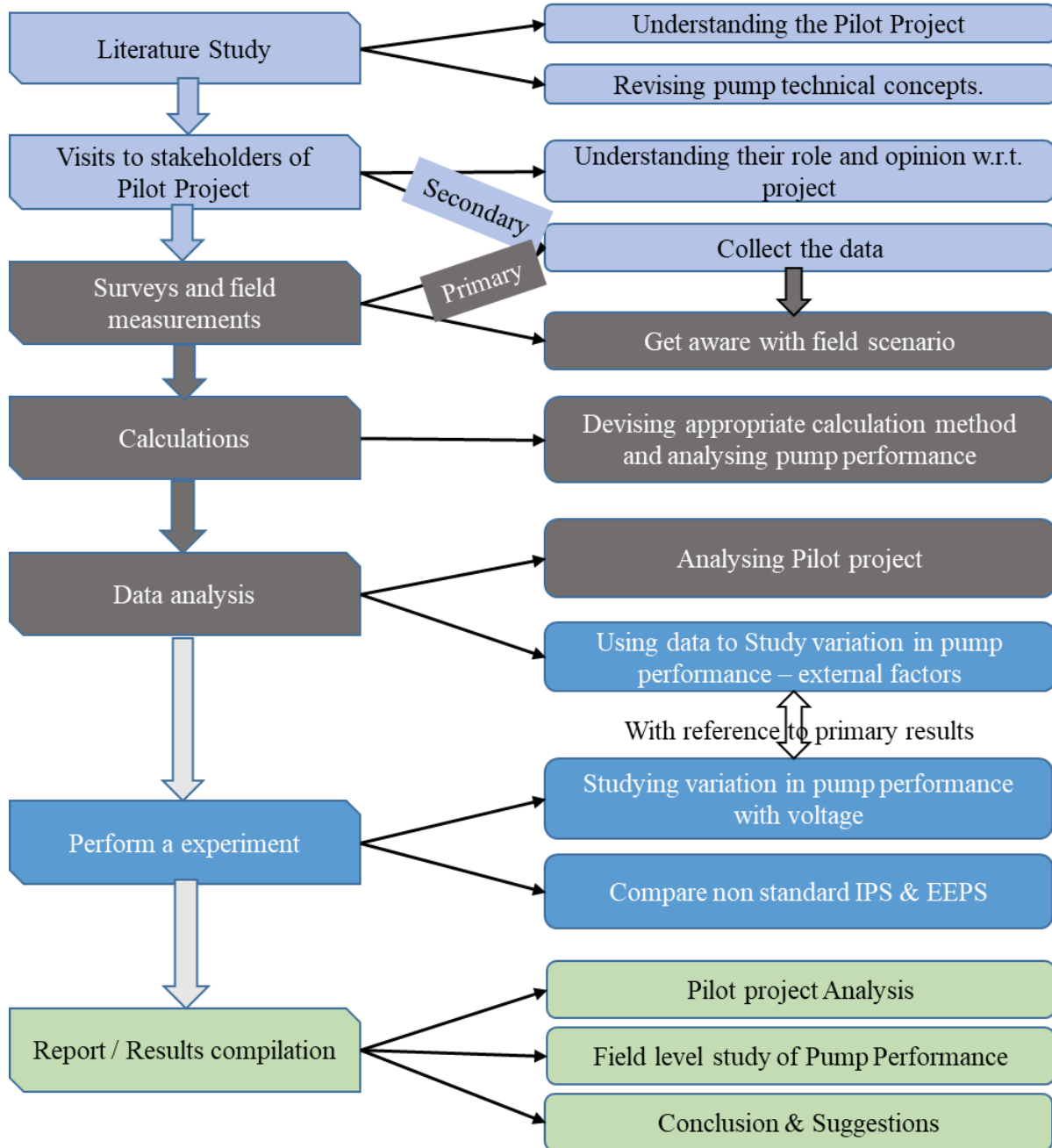


Figure 1.4 Schematic of work overview

1.5 Data Collected

1.5.1 Primary Dataset:

For collection of primary data about 27 of beneficiaries were visited. At few location only measurements were not possible while at few beneficiary weren't available during survey. Hence, 22 out of 27 of good results were recorded. The note to be taken that all datpoints used for technical measurments and qulitative observations are not the same, few does not have one or another record.

Table 1.1 List of primary data

List	Description	Source of origin
Technical measurments	Measurements of IPS installed under Pump replacement project. It includes: Pump performace measurements.	Field survey
Qualitative observations	Field observations for condition of IPS, Farmer responces. Responces of 22 beneficiaries.	Field survey

1.5.2 Secondary Dataset:

Secondary data can be useful to do the primary analysis and establish baseline. In addition to this, as pilot project has just concluded the most recent data can be obtained for accurate analysis of project. Following is the list of data obtained by visiting the concerned stakeholders.

The data is used for M&V analysis of pilot project and studying the performance of IPS in the fields.

Table 1.2 List of secondary data

List	Description	Source of origin												
1 st & 2 nd M&V measurement of Mangalwedha pump replacement project.	This contains detail measurements taken during M&V and installation of IPS, and calculations. Following are the contents of the data. <table><tr><td>430 IPS measurements</td></tr><tr><td>Piping details (L, D)</td></tr><tr><td>Discharge</td></tr><tr><td>Heads required for calculations</td></tr><tr><td>Consumer name, feeder, village etc.</td></tr><tr><td>Pump type, water source</td></tr><tr><td>Input power measurements</td></tr><tr><td>Pump model and ratings</td></tr><tr><td>Installation date</td></tr><tr><td>Meter no. and reading for few</td></tr><tr><td>Saving analysis</td></tr><tr><td>Calculated efficiency</td></tr></table>	430 IPS measurements	Piping details (L, D)	Discharge	Heads required for calculations	Consumer name, feeder, village etc.	Pump type, water source	Input power measurements	Pump model and ratings	Installation date	Meter no. and reading for few	Saving analysis	Calculated efficiency	MITCON Pune, (M&V Agency for the Project.)
430 IPS measurements														
Piping details (L, D)														
Discharge														
Heads required for calculations														
Consumer name, feeder, village etc.														
Pump type, water source														
Input power measurements														
Pump model and ratings														
Installation date														
Meter no. and reading for few														
Saving analysis														
Calculated efficiency														

List of EEPS came to repair during the project duration.	A free repair and maintenance was provided under the Project; for the duration of 5 years from the date of installation. Following are the contents: List of 2567 repairs Beneficiary details – Name CRI ID, Feeder. Pump details – Model, Type, HP rating Repair details – Date, repairs.	CRI project office, Mangalwedha
Cost benefit analysis	Briefs of actual financial benefits from Project.	MSEDCL Mumbai
Feeder energy consumption in mangalwedha	Monthly energy consumption on Borale 11kV agricultural feeder for the year of 2016. Annual Energy consumption for feeders which was part of pilot project.	Respective substation, MSEDCL, Mangalwedha
Agricultural load on feeders under Project.	Total number of IPS, and sanctioned load on 4 feeders on which EEPS has been installed.	MSEDCL, Mangalwedha
Pump Catalogues	Head and discharge values for particular Pump.	Internet

Chapter 2 LITERATURE REVIEW

2.1 Efficiency Scenario in the India

(Reidhead, 2001) involves a study stating the field scenario of energy efficiency of irrigation pumpsets and analysis of various measure for improving efficiencies. Discussed measured includes comparative analysis of different rectification measures with a protocol to choose an optimum one. (Patel, 1988) If pumpsets operated perfectly the observed efficiencies of pumpset can be found as low as 13% for electrically operated while 5% for diesel operated pumpsets. Author has finely linked the end user to the performance variation of IPS. Proper pump selection, operation and upkeep are found to be major root cause for low performance. According to author from government's point of view the rectification measure is logical if subsidies are removed but will be politically costlier. In brief, apart from engineering analysis financial discretion and appropriate implications of policies and institutions has to be considered to achieve sustainable energy savings from agriculture sectors.

(Bhatiani, G & Goyal, R, 2010) has gathers results for power saving potential with different end user rectification measures. Usage of efficient pumps has been considered as to have maximum energy saving potential of about 25-40% as shown in following table. Continuous awareness building of farmers, capacity building of farmers as well as ESCOs, usage of technology to achieve water efficacy, incentives through DSM and initialisation need to be focused for sustainable performance of irrigation in the country.

Table 2.1 Energy saving potential with different rectification measures

S.N.	Measures	Potential for Savings
1	Efficient Pumpset	25-40%
2	Suction Pipe & Foot Valve	5-15%
3	Efficient Irrigation Methods	15-25%
4	Total Saving Potential	45-80%

2.1.1 Role of Power supply utilities:

Vast scale DSM program execution in India require huge specialized and additionally money related venture with extensive monitoring and administration framework. The net advantages of

DSM are the negligible cost of energy and its sale to the consumers. (CRISIL, 2012) Utilities prefer short term gain over the long term benefit by investing in demand side management; As they have a financial burden of salaries of employees, debts, asset maintenance etc. (Abhyankar N, Phadke A. 2012) Because of uncertainty and irregularity in power demand, fuel prices, load shedding etc It becomes challenging for the utilities to monitor and verify the savings generating from the DSM activities. (Ahn S, Graczyk D. 2012) Awareness programs are one of least complex activity to conduct under DSM. But, Planning and execution of such a fundamental DSM program costs the utility. The cost and resulting outcomes does not necessarily have sufficient incitement to utilities for doing such measures. In addition to such a low net benefit, there are no legal bindings for utilities to implement such DSM activities in India.

(Harish, V. S. K. V., & Kumar, A. 2014) states majority of DISCOMs in India are possessed by corresponding state governments. There is always pressure to keep the tariffs low for politically influenced sectors like agriculture. The revenue collection efficiency from agriculture sector is very low. As a result, majority of DISCOMs undergo heavy losses. This creates the scarcity of funds restricting the upgradation of distribution network etc. And then state govt. has to look for the support from central government for finances.

2.2 Overview of past AgDSM Projects:

Table 2.2 provide the brief details of few major past AgDSM projects, implemented across India. Projects involving replacement of pump accessories has been widely implemented. While very few projects dealt with improving the power supply quality by infrastructure development etc. In Mangalwedha project with replacement with EEPS 5-year free maintenance is also provided, which will make sure the sustenance of pump performance, this is a major characteristic of this pilot project.

Table 2.2 Overview of Past AgDSM Projects

Sr. No.	Project Location & Period & IA	Activities	Outcomes
	Rectification with only Accessories replacement		

1	Gujarat - 1978-88 - GEB	Replacement of suction pipes and foot valves for 1108 pumpsets	21.7% energy saving.
2	Madhya Pradesh - 1980 - MPEB	Replacement of suction & delivery pipes and foot valves for 3125 pumpsets.	(527 case evaluation reported - Energy consumption increased by 12%, System efficiency improved from 23-33%
3	Andhra Pradesh - 1985-86 - APSEB	Replacement of suction & delivery pipes and foot valves for 2500 pumpsets in 2nd phase..	(505 case evaluation reported - Energy consumption increased by 2%, estimated energy savings was about 26%.
4	Maharashtra - 1989-94 - MSEB/MEDA	Replacement of suction & delivery pipes and foot valves. About 350-400 IPS were rectified per year.	No M&V particulars available.
5	Tamilnadu -1991-92 - MoP	Pump rectification by Ministry of Power	Target 10000 IPS
6	Punjab - 1997-98 - MoP		Target 1350 IPS
7	Haryana - 1997-98 - MoP		Target 800 IPS
8	Gujrat - 1997-98 - MoP		Target 5000 IPS
9	Andhra Pradesh - 1996-99 - MoP		Target 20000 IPS
	Rectification with Pumpset & Accessories replacement		

10	Tamilnadu - 1980 - TNEB	For 5 HP IPS, suction piping, delivery piping , foot-valves were replaced for about 3125 pumpsets.	Average of 19% energy savings reported. while 11.7% increased energy consumption due to increased discharge.
11	Gujrat - 1985-86 - GEB	Replaced 5 HP IPS, foot-valves were replaced for about 3128 pumpsets.	Estimated 34.7% energy savings, Evaluation of 527 cases shoed 48% reduction in energy per unit of water discharge
Rectification and Energy efficient pumps			
12	Madhya Pradesh - 2004 - Kirloskar/CIDA/Econol	Replacement of Old pumps with efficient IPS. 50 replaced.	Examples of two cases indicate energy saving of 33.8% and 43%
13	Gujarat - 2007-09 - GEDA	Replacement of Old pumps with efficient IPS. Submersible IPS replaced with BEE rated 5 HP IPS , 6000 replaced.	No data
14	Karantaka (Doddaballapur) - 2006-08 - WENEXA/USAID	Replaced old pumps with accurately sized efficient ones with condition to use drip irrigation. 15	Reported about 70% decrease in energy and 60% in water consumption. (data was

		farmers participated.	self-reported by farmers)
15	Karnataka (Malavalli) – 2014-19 – CESE/EESL	Replacement of Old inefficient pumps and accessories with EEPS, and free repair for project duration.	Expected power saving of 37% annually.
	Promotional Activities - promoting standard pumps, Demonstrations etc.		
16	Gujrat - 1986 - GEB & Kirloskar Brothers Ltd.	Demonstration of pump replacements at 16 farms, witnessed by about 700 farmers;	Initial energy saving 24 - 69%; Total energy reduction of 53% during rectification
17	Haryana - 2000 - HAREDA	Promoted the use of ISI marked pumps with some financial aid. (till 2009, 14668 ISI standard IPS installed)	Reported savings of 46.6MU, methodology not clear.
18	Karnataka -2009-10	Demonstration project, Replaced old pumps with EEPS with metering for 50 Farmers	Metered electricity use fell by 41.5% with 22% reduction in water usage for beneficiaries.
	Utility measures - Power supply side improvements		
19	Andhra Pradesh (Nalgonda) - 1998-2000 - APTRANSCO	Conversion of existing 3-phase LVDS into high voltage DS. Installed single phase pumpsets instead of 3-phase (target	Efficiency of pumping improved from 27 to 36%. Scheme ended prematurely

		3216 pumps more than 1641 achieved)	
20	Karantaka (Doddaballapur) - 2009 - BESCOM/ENZEN	Transformation of pilot feeders to high voltage distribution system (HVDS) with one distribution transformer (DT) for each pump, installation of capacitor banks, fault passage indicators, and fixing of meters on all DTs and is responsible for maintaining the power supply system.	665 DT Installations done one for each pump
21	Karnataka (Bantwal) - 2003 - Co-op Society	Replace 13143 inefficient IPS in taluka with EEPS	failed as Co-society did not receive distribution rights in region.

(Source: D'Sa, A. 2010, Prayas Pune)

2.3 Scope of Efficiency Improvement:

(Sant,G., & Dixit, S., 1996) in role of standards in agricultural pumping efficiency has analysed, how standards can help achieving the improved the pumping efficiency in agriculture. Author has linked the filed scenario, farmer perception and situation of standards for pumps in India, giving a broad understanding into dynamics of the topic. There is difference with NABARD and BIS standards generating unnecessary confusion filed implementation. In addition to this the standards are old and very few amendments have been done, there is need for revision of

standards considering the present field scenario and technological advancements. While interlinking the significance of standard with field author states, problem of proper pump selection which is one of major root cause of low efficiency can only be solved by improved awareness and education of farmers but, IPS with better characteristics which can be achieved with defining proper standards will definitely perform better and give improved efficiency taking care of improper selection at some extent.

(Bredin, M. S.,2017) has investigated the different ways of improving the pumping efficiency. With pumping simulation for few case studies, Bredin obtained results for energy saving potential by different methods. Servicing the pump and trimming the impeller can save up to 15%, replacing the pump and motor, can save 20%. While Installation of VSD with feedback control system can save up to 6%. Though the saving values will vary with location and overall pump usage pattern, above analysis gives the baseline of approximate achievable savings. In addition to this a methodology that can be used.

(Saidur, R. ,2010) carried out study on improving the efficiency of electric motors and found that the most motors are efficient if operated above 75% load. Author studied the different losses in motors and energy saving potential. These methods are divided into three parts one Using regulation like awareness and education, enforcement of standard etc, second technology application like usage of capacitor & VSD; third housekeeping like maintenance, energy auditing. The motor failure rate due to contamination is about 20% while due to overloading is 25%. The study regarding usage of VSD has been carried out. VSD is an option to improve efficiency but not economical one. (Capehart, B. L., Turner, W. C., & Kennedy, W. J., 2006) found that Motors lubrication can save about 1–2% energy.

(Kini, P.G., Bansal, R.C., 2010) carried out the study of voltage and load variation on the performance of centrifugal pumpset. Efficiencies of Motor, pump and system all has been studied with respect to balanced and unbalanced voltages. Results shows that efficiency of motor decreases with decrease of slip, while efficiency of pump increases and then decreases considerably at balanced voltage conditions. Over the balanced voltage condition across three phase the pumps efficiency found to be dominant in overall system efficiency, suggesting change

in pump design for efficiency improvement. Though the author has proved the importance of consideration of voltage fluctuation on the pump performance, the study is based on experiment of single pumpset. It does not discuss about application of pump, linking it to a practical scenario. Hence a field scenario or application connect is need to be established to study the actual possible variations. Also, a simplified study with different voltage and loads instead of being in terms of balanced ones will improve the understanding of the subject.

2.4 About Pilot Project:

BEE launched a pump replacement pilot project under AgDSM scheme in year 2009 in Solapur, Maharashtra. The Mangalwedha block has been selected for implementation of pilot project. A Kharatwadi feeder in Pandharpur block has also been part on the project. Under this project the old inefficient pumps in Mangalwedha block were replaced and the free maintenance was provided for the period of 5 years. Four agricultural separated feeders namely Bhose, Borale, Nandeshwar & Brahmapuri selected for pilot Ag DSM project. (MITCON, 2017)

Maharashtra State Electricity Distribution Company Limited (MSEDCL) supplies power to the agriculture pump all over Maharashtra and played the role of facilitating agency for this project. Company CRI pumps was implementing agency which carried out the all the work with the help of their authorized local distributor. MITCON Consultancy & Engineering Services Ltd., (MITCON) is the third party Monitoring & Verification agency appointed by Bureau of Energy Efficiency (BEE) and supervising the Ag DSM project. Monitoring and Verification (M&V) is

the process of using measurement to reliably determine actual savings created within a project by an energy management program. (MITCON, 2017)

The project duration was 5 years and has ended in Dec. 2017. Initially target was to replace 3530 pumpsets under this project, but due to field level difficulties total of 2209 pumps was replaced. The project implementation has been done on the basis of hybrid ESCO model, in which complete investment was done by CRI Pumps Ltd. and monetary benefits from achieved energy savings were shared between CRI & MSEDCL on a 30:70 ratios respectively.

The baseline energy consumption was estimated based on average of power input of old pumps tested before replacement with calculated 1640 hours of annual operation. And these hours of pumping will be considered for energy saving calculation for the project period of 5 years. To calculate the energy saving, minimum 10% of replaced pumps was selected randomly for testing, and every year different pumps are tested. The energy savings are priced at Rs. 2.70/unit for a project period of five years. The proposed project budget was approximately Rs. 442 Lakh which increased to about Rs. 839.42 Lakh, as the cost of pump accessories that need to be replaced with pump was not considered as a part of DPR preparation. Apart from the revenue by energy savings additional Rs. 8.8 lakh per year was given by MSEDCL to IA (i.e.,CRI Pumps) for operation and maintenance of project from LMC funds. (MITCON, 2017)

Chapter 3 Current Status of Project Location

3.1 Brief Overview of location:

Solapur is 4th largest district in Maharashtra. District has dry (arid and semiarid) climate. Climate here is characterized by high humidity through year, an oppressive summer followed by well distributed and heavy rainfall during the monsoon with an annual average rainfall of 625mm. The major Rivers flowing in the district are the Man, The Bhargavati, The Bhima, and The Sina Rivers. About 26% of total cultivable area is cultivable (i.e.,2961 sq. km.out of 11480 sq. km.). The major soil type is derivative of basalt rock, locally known as Murum. The pre-monsoon depth to

water level ranges from ground level to 12.80 m while post monsoon water level ranges from 1.00 to 19.60 m below ground level. Jowar is major crop taken in the district with contribution of 70%, while wheat, gram, corn & sugarcane are other major crops. ((CGWB, 2013) (Feedback, 2010) (GSDA, 2012))

Mangalwedha is a taluka in Solapur district, selected for pilot project of AgDSM. Jowar is a major crop in the area, while sugarcane is taken depending upon water availability and dominant in northern region of Mangalwedha along the river side. The average annual rainfall in the block is about 510mm according to 2011 figure. The average annual fluctuation in groundwater is about 3m with pre monsoon average level being 7.85mbgl and post monsoon being 4.22 mbgl. Following figure shows the location of feeders where Pilot project has been implemented. A feeder in Pandharpur block has also been selected for pilot project which is not shown here, as it is not considered in this study.

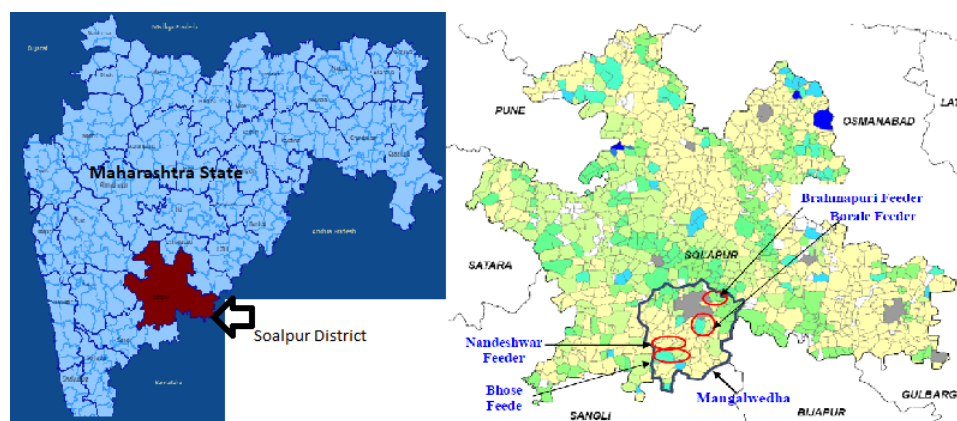


Figure 3.1 Pilot Project map – Mangalwedha

(Source: DPR 2009)

3.2 Scenario of irrigation among Surveyed beneficiaries

To know the perception of farmer towards irrigation Pumpsets and pump replacement pilot project in Mangalwedha, a survey has been conducted of 22 farmers. The detailed questionnaire is in appendix. In addition to this, few of responses of farmers are objectified for analysis, data can be found in annexure. The farmers surveyed were the beneficiaries of pilot project.

Figure 3.2 shows the distribution of types of Pumpsets surveyed. From this we can see that the dug wells are the major source for irrigation in surveyed area. The average power rating for sample is 5.52 HP. It seemed that submersible Pumpsets are comparatively more used. The reason behind this can be they need low maintenance, for monoblock one has to be cautious about keeping it dry, proper coupling etc.

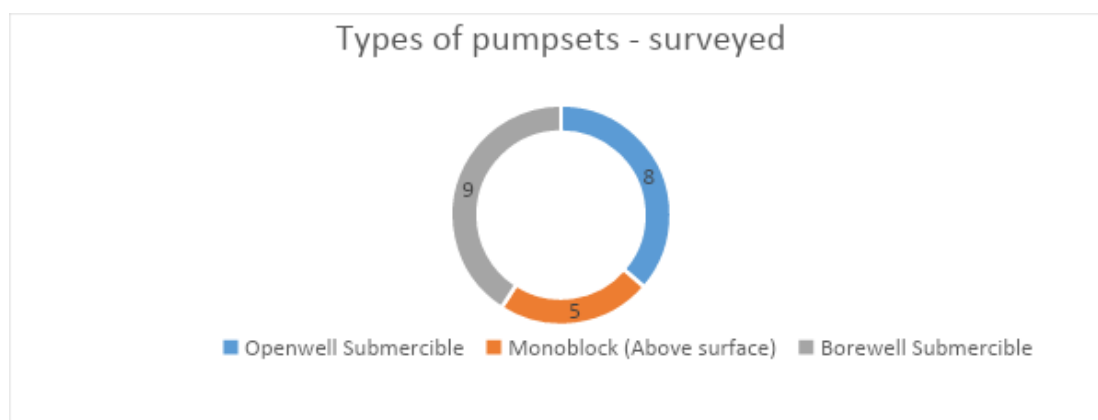


Figure 3.2 Distribution of various types of IPS surveyed

Table 3.1 briefly tells us about the pattern in farmer behaviour regarding irrigation usage. As we can see, with 78% of farmers owning more than 2ha land, 67% of them with too old pumps, very few i.e., 18% with ineffective irrigation technique and 50% with over usage, this delineates the low performance irrigation scenario in the area. About 67% of surveyed farmers were using pump older than 5 years at the time of pump replacement project. As pumps get older; its performance is likely to decrease. Since new pump has been provided to farmers; majority of them has been benefited with a better performance pumpset at right time. Again, about 50% use autostarter which implies (and confirmed with cross questioning) continuous usage of pumpset till the power is available i.e, about for 8 hrs in a day. This implies ineffective operation of pumpset.

Table 3.1 Irrigation scenario among beneficiaries

Content	%
Pumpsets (replaced ones) older than 5 years	67

Beneficiaries with irrigation area more than 2 ha	78
Drip users	18
Farmers using Autostarter (continuous operation 8hrs)	50

Installation of energy meters is mandatory for accessing the electricity through DISCOMs. With the energy meters tracking of energy consumption becomes easy. Through the Pump replacement scheme new energy meters has been installed for every beneficiary. During survey it has found that after 5 years of installation only about 18% of surveyed beneficiaries had the meters in working conditions. While more than 50% had meters installed but were not in working condition. Figure 3.3 represents the condition of meters on field. With regard to this scheme, as benefits of schemes are measured based on energy consumption data, presence of working meters would have played an important role. A deeper investigation is need to be done to find the reason behind failed utilization of meters on field.

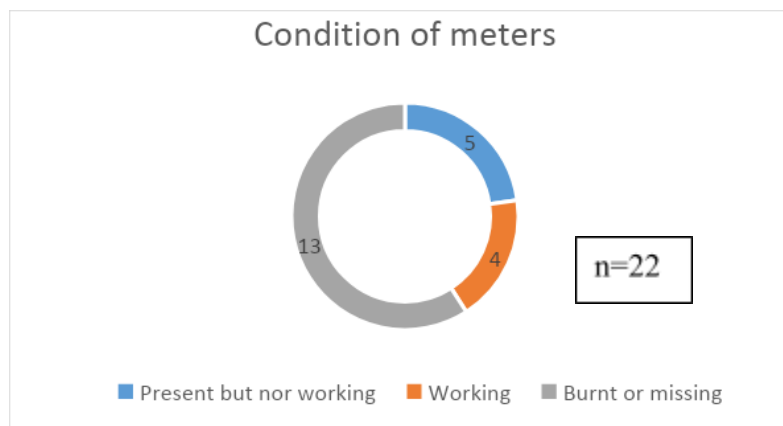


Figure 3.3 Condition of Energy Meters of beneficiaries

3.3 Observations about M&V of the scheme:

MITCON PVT LTD. was the 3rd party M&V agency for this pilot project. Primary objective of the agency was to measure the efficiency of 10% of total installations and calculate the energy savings. Hence the results of M&V would have been the measure of success of the project. Any error during measurements and/or calculation would lead to wrong judgement about the performance of the project.

Such M&V data is not only being used to measure performance of project but to do various technical study related to whole pumping system as well. It can be used to find out the linkages between various factors like breakdown, time, voltage efficiency etc. depending upon the contents of data. Hence accurate measurement and calculation of data has far more importance as it can help us to take decisions for further improvements.

The measurements for M&V are taken yearly generally during December to February. To measure the efficiency of IPS myself, I roamed with the person appointed to take measurements during their 5th year M&V. Following are few inferences drawn based on observations on the field. This may help in improving the M&V work in future project.

One of critical observation is related to tapping point for discharge measurement. It has observed that the discharge was measured at convenient point rather than at the end of pipe line/ last chamber. While the length of pipeline was asked to the farmer which itself may possess some error and that to a complete length of pipe line was recorded where as it should have noted till the tapping point. For longer pipes this would actually result into greater frictional head if distance between tapping point and pipeline end is large.

Another possible error that was found is in the calculation of friction head. The frictional head is a very sensitive term, the error in discharge, length or friction factor measurement can cause the aberrant results. In this case we found the friction factor taken to be very low compared to what we have calculated. This has resulted into reduced frictional head compared to original one.

In addition to this, even if some farmer connects a different pump (at the place of pump replaced under scheme which need to be monitored), in case of submersible it is very less likely to know about it, this may affect the performance evaluation of the scheme.

Chapter 4

Investigating broader reasons for efficacy of scheme

4.1 Issues due to maintenance

Figure 4.1 represents the broader problems affecting performance of IPS in fields. It is observed that about 52% of time the problems are caused due to farmer misuse. (Priyanka, 2016) In addition to this there was severe draught during the initial 3 years of project duration. This translated into many problems due to low water levels. To study this in depth, an exercise involving farmer misuse which is mainly characterised by few maintenance checks and breakdown of pump is carried out.

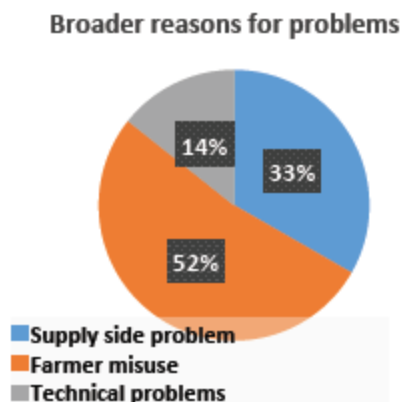


Figure 4.1 Broder reasons for inefficient IPS

(Source: Priyanka, 2016)

To find out if there is any correlation between farmer behaviour (here only in terms of pump maintenance) and the pump breakdown, data collected by survey is used. Sampled beneficiaries

from surveyed population, experienced the breakdown of pumpset with their older IPS as well as new EEPS. As majority of pump breakdown are associated with Electrical parameters, electrical panel check has been observed to verify the correlation.

Based on observations in filed during surveys, a condition check for electric panel is carried out as it can be linked with farmer misuse. Figure 4.2 shows the summary of checks performed. How the checks are defined as poor is mentioned below:

For Fuse: Missing, Filament used is inferior, signs of fuse burst.

Meter : Missing, not working, Signs of meter burst (black spots).

Starter : Missing, Protection panel removed and tampering of default settings, need manual tripping (not working properly).

Presence of these components on electrical panel is not only important from point of view of performance of IPS but are essential for safety of human. From this we can conclude that, proportion of misuse by farmers is verified. It is substantially poorer than standard conditions and need to be improved in order to improve effectiveness of energized irrigation.

In further sections, we will be looking if there is any correlation between maintenance and breakdown of pumps.

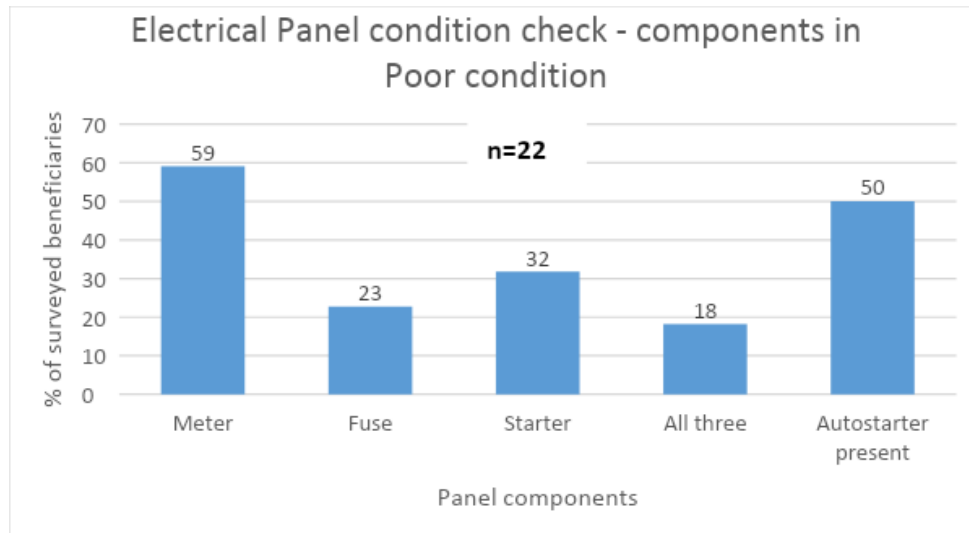


Figure 4.2 Electrical Panel condition check

4.1.1 Pump Usage & Breakdown Interlinkage:

Table 4.1 indicated the condition of electrical panel as observed during field survey. Here, farmer maintenance is associated with the condition of panel and breakdown. To see if farmer misuse/maintenance affect the pumpset. Reading 1-4 are for the IPS which broke down while 5-8 are randomly selected from bunch of surveyed IPS which never broke down. Here Bad condition is referred to the scenario which should not be present according standard condition e.g., fuse must not be bypassed or improper filament should not have been used etc. From this exercises it is evident that a valid correlation cannot be established between maintenance of IPS and its breakdown. There are many such factors which affect such scenarios, like voltage, timely on-off, weather etc. Hence finding the correlation and establishing the causality becomes difficult.

As we can see in majority of cases where auto starter is used that is pump is operated automatically when power comes and goes, the breakdown has occurred. Similarly, absence of auto starter shows the breakdown in minimal cases. And if pumps are not breaking down without auto starter suggests the farmer/beneficiary pays an attention to its regular operation (maintains it) or not using at all (which is not the case here). Sudden shutdown of pump at power cut, frequently on-off due to irregular power, and continuous (more) usage can be the reason for breakdowns for beneficiaries who are using autostarter.

Table 4.1 Linkage between Breakdown & Maintenance/usage

	IPS which broke down				IPS which never broke down			
	1	2	3	4	5	6	7	8
Condition of starter	Good	Bad	Good	Good	good	Good	Good	Bad
Condition of Fuse	Good	Bad	Good	Good	good	Good	Good	Good
Do you use Autostarter (continuous operation 8hrs)	Y	Y	N	Y	N	Y	N	N

(Vasudevan, R., Cherail, K., Bhatia, R., & Jayaram, N., 2011) found that, power utilities distribute the power in rotation to consumers, it is rotated between farmers in 2 slabs of 4-8 hrs per day. But this has led farmers into leaving motor switches in on position to pump the water whenever power is obtainable. The starting current is very large resulting in large voltage drop when the power comes, resulting damage. In addition to this, a large number of pumps got activated at the same period, which increased the load demand and often resulting burning out the transformers and IPS.

This proves that over usage of IPS or rather usage of auto starter can be one of reason leading to IPS breakdown, which can be accounted under farmer misuse.

4.2 Investigation of the Pump Selection:

Selection of appropriate pumpset for the field is important, as it directly affects the performance of complete ecosystem around it. Though there is no single answer about how farmer select pumps, generally selection is based on knowledge gained by word of mouth for said HP of pump or suggested by pump dealer. Also, making farmers aware about efficient pumps and proper selection was part of pilot project. Hence to know if beneficiaries has learned about it this investigation is made.

To investigate this, few parameters were selected which farmers desire in a pump. During survey farmers were asked to prioritize the parameter for process of pump selection. Finding out farmer awareness level is a difficult task. From field survey I noticed that, though farmer may not be aware about all technical details related to Pumpsets but they know what they want; based on

which they make their decision. To verify this, I tried to find if they are aware about the head and discharge. Head (pressure head) and Discharge are two important parameters while selection of pumps. finding head and discharge itself can be a difficult task, usage of these terminologies for pump selection shows the concern about selection of appropriate pumps in some ways.

4.2.1 Consideration of Head & Discharge for pump selection

Figure 4.3 represents the awareness of farmers among surveyed population. It is found that about 61% of respondent consider head and discharge required while making a pump selection. Most of them may just be taking the help of pump dealer or technician for figuring out the required head and discharge. They are at least aware about the terminology and knows that it is going to affect the performance of their pumping system. This concern will not optimize the performance of whole pumping system completely; it brings one step closer to usage of optimized system.

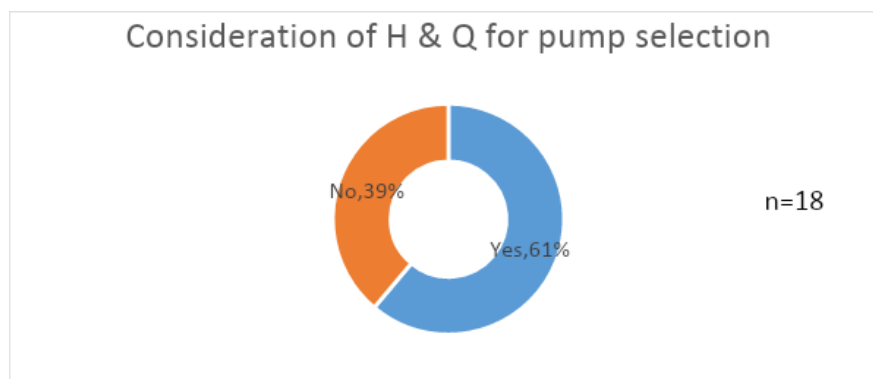


Figure 4.3 Consideration of H & Q for pump selection

Now linking this with the Pump replacement scheme, implementation of this scheme has introduced them to new pumps, the performance of which are far better than non-standard Pumpsets. Now that they are acquainted with better performance Pumpsets the level of awareness about optimized pump usage is expected to increase.

4.2.2 Priority Ranking:

Table 4.2 shows the distribution of surveyed population and their priorities to pump selection. It shows the percentage of surveyed sample ranking the particular parameter. E.g., column '1', 53% of respondents have ranked low voltage compatibility of pumpset at topmost.

Voltage compatibility is referred to the pumpset that can function properly even at lower voltage (generally between 250-380 volt line voltage). While the voltage that need to be maintained ranges from 377-340 Volts. (-9% to +6% of 415 Volts)

Good quality/ standard is referred to pumpset which are rigid/sturdy, and complied to some standards. Low breakdown frequency.

Durability is defined by the lifecycle of pumpset.

Guarantee: Few manufacturers provide guarantee i.e., assurance of pump performance for particular time period.

Cost: It is the market price associated with product. Initial investment.

Free Service: Work related to pump installation, repair, maintenance which are free of cost.

Table 4.2 Distribution of respondents to priority ranking

	% of respondents for Respective rank/priority					
Parameters/Ranking	1	2	3	4	5	6
Voltage compatibility	53	18	24	0	6	0
Good quality/standard	24	24	6	24	24	0
Durability	18	41	35	6	0	0
Guarantee	6	6	6	35	47	0
Cost	0	12	29	35	24	0
Free service	0	0	0	0	0	100

Figure 4.4 shows the priorities of respondents for pump selection. As we can see voltage compatibility has been ranked 1st and hence the most important parameter. It is observed that low voltage compatibility of IPS, standardized IPS and durability are most important parameters for

farmers. As these Three parameters has always been in top three. Free service has been of lowest priority for all the respondents, which was thought to be a benefit from farmer's point of view under the pump replacement scheme. Cost seems to occupy the fourth position in this ranking.

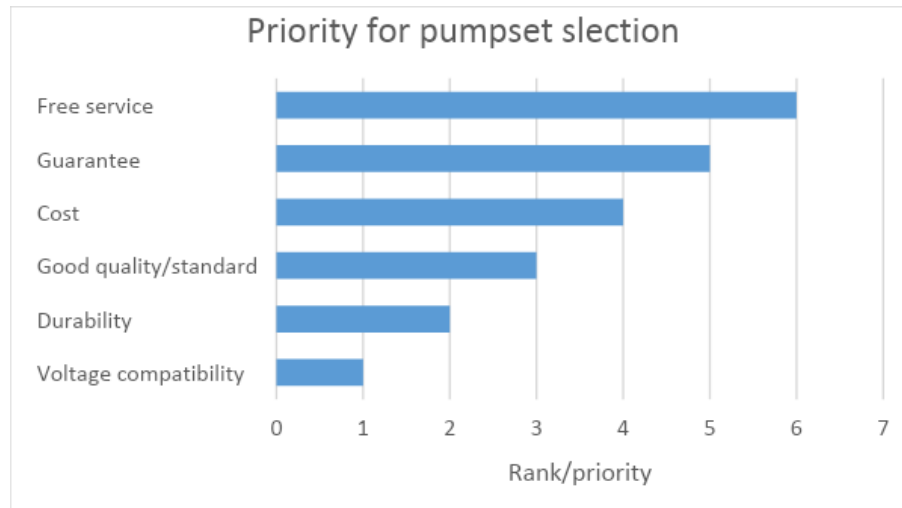


Figure 4.4 Pump selection Priority ranking among beneficiaries

4.3 Issues due to Quality of power supply

Power quality is one of the critical parameter which affects the AgDSM schemes. Efficiency of IPS and its breakdown were vital components of ESCO project. As both of these factors are directly affected by quality of power supply; it is expected to have good quality power supply for achieving desired energy savings. Now, if we look at the data, we can observe that the avg. pump rating at most of the feeder is more than sanctioned, even after project implementation. This may lead to overloading and become reason for unreliable power supply. Also, voltage fluctuation a one of the major cause for pump-motor damage, ensuring proper supply within permissible limit of about $\pm 10\%$ of required can help in achieve better efficiency.

Table 4.3 Utility side issue – Sanctioned HP vs Actual HP

Feeder Name	EEPS Nos.	Sanctioned HP	Old Pump	New Pump
Old pumps Data		HP	HP	HP
Bhose	339	4.4	5.1	4.6
Borale	672	4.2	4.9	4.4
Brahmapuri	266	5.9	6	5.8
Nandeshwar	500	4.7	5.9	5.2
Kharatwadi	432	5	5.7	5.2
Total	2209			
Average		4.7	5.5	4.9

Considering this in mind, quality of power has been checked during survey. Here the good quality power supply is referred to the line voltage above 350 volts at the respective points. Farmers were asked if they face issue of low voltage regularly, which is verified with instantaneous measurements. Based on these the samples are categorised as good and poor quality of power supply. As seen from figure 4.5, approximately 40% of surveyed population reported poor quality of power supply. Based on common observations on field, electricity theft can be one of the reason. As utilities design infrastructure for particular load, proportionality of electricity theft is one things that cause imbalance in load management, leading to poor quality of power supply at desired location.

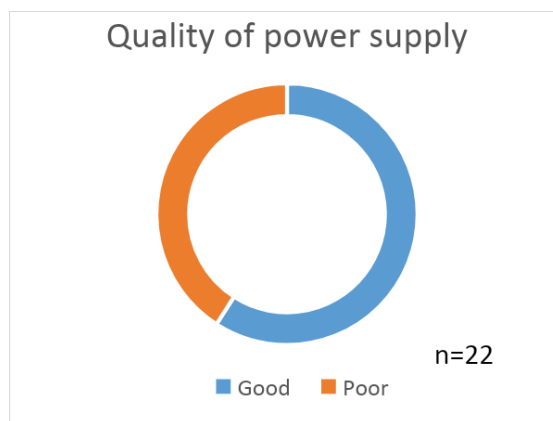


Figure 4.5 Quality of power supply

Chapter 5 Investigating the Hours of Pumping

5.1 Background

Hours of pumping is a crucial factor while discussing the performance of irrigation system. It has direct link calculation of in energy consumption and hence the payback of scheme. Few methods of determining hours of pumping are considering; Metered consumers, cropping pattern, finding energy consumption with feeder analysis etc.

For pump replacement project in Mangalwedha, annual hours of pumping were calculated in DPR as 1640 and assumed to be constant for next 5 years of project duration. As financial benefit of project was calculated based on hours of pumping any variation in this number would have drastically affected the project positively or negatively. Keeping this in mind, hours of pumping have been investigated. In this section I am going to discuss different findings about pumping hours in Mangalwedha block.

5.2 Methodology adopted by pilot project:

While calculating the hours of pumping for DPR; two approach has taken. Under first approach annual average of past 3 year's energy consumption of metered consumer on same feeder lines under pilot project has been used to estimate baseline energy consumption. Under Second approach existing pump sets has been audited to estimate baseline energy consumption of pilot

project. By analysing the field study measurements average operating efficiency and average input power in kW are estimated, for existing pump sets of different types such as monoblock, submersible and flexible coupling and for different HP ratings. This average energy efficiency and average input power norms along with assumptions of average operating hours has been applied to total number of pumpsets. Which were categorised as per their ratings and types to arrive at baseline energy consumption. Based on annual average of past three year's energy consumption figure of 21.16 million units taken at feeder level recorded at the substation end. And considering the distribution losses of 26.2% the final energy consumption for 2221 pump sets is estimated to be 15.62 million units. However, baseline energy consumption obtained by approach 1 is 16.49 MU. Based on energy audit study, the sum of average input power for all the pumpsets was around 9523 kW. Considering the conservative situation least of these values has measured. (DPR, 2009)

$$\text{Annual Average Operating Hours} = \frac{\text{Energy Consumption, } 15.62 \text{ MU} * 10^6}{\text{Sum of average input power for all the pump sets, } 9523 \text{ kW}} = 1640$$

$$\text{Annual Average Operating Hours} = \frac{\text{Energy Consumption, } 16.49 \text{ MU} * 10^6}{\text{Sum of average input power for all the pump sets, } 9523 \text{ kW}} = 1732$$

(DPR, 2009)

5.3 Analysis:

In approach 1, energy consumption for all metered connection is added and then it is divided by total pump rating which included IPS with 3hp rating to 40hp. But as number of high capacity IPS i.e., >20hp was very less and weightage average has not been considered, the results obtained for lower rating pumpsets are comparatively different, which has discussed in succeeding subsections. While, in approach 2 average energy consumption for year 2006-09 is considered. As in this year the hours of pumping being more than 10 hours which reduced to 8 hours after year 2009 the results show little variation. Also the consumption in 2006-07 is

higher compared to year 2008-09 increasing the average and hence the hours of pumping the detailed calculation is discussed in succeeding sections.

5.3.1 Scenario 1: Feeder level

In this case I have used the same approach 2, used in DPR, but just for Borale feeder. Borale Agri. Feeder has about 1/3 rd of total energy consumption among all 5 feeders and 30% of total pumps replacements. The 26.2% loss is considered as it has been used in DPR to maintain similarity between approach 2. It has observed that the calculated annual hours of pumping are coming out to be less than 1000 for years before and 4th year of project. Even considering the average power consumption for Borale feeder the consumption is very less compared to what is considered for the project. Though Borale feeder amounts majorly among all 5 feeders, it can be said that there is quite an imbalance of energy consumption, load and hence hours of pumping among 5 feeders.

Table 5.1 Hours of pumping details - based on Feeder readings

Sr. No.	Year	Energy consumption kWhr (A)	Assuming 26.2% losses (A*0.738)	No. of pumps (B)	Avg. Measured Power rating (kW) (D)	Total Ag Load (B*D)	Hrs of pumping (A*0.738)/(B*D)
1	2006-09	6270778	4627834	682	5.83	3976.06	1164
2	2008-09	5135038	3789658	682	5.83	3976.06	953
3	2016	3246000	2395548	600	4.94	2966.75	807

(Source: 1st and 2nd reading from (DPR, 2009); 3rd Borale feeder records)

5.3.2 Scenario 2: Metered

During M&V energy meter readings has to be recorded. During field visits it has observed that very few meters were in working conditions. Data shown in Table 5.2 is imported from 1st M&V detailed measurement report. It has observed that out of 308 measurements done for this particular M&V, only for 101 meter readings were recorded. Avg. Meter reading is the average of energy consumption measured at 101 beneficiaries. Average number of days is the difference between date of installation of IPS and approximately assumed date of measurement. As

measurements go on for about 2 months I have considered the date in between, to reduce the error. Based on energy consumption and no. of days of operation

Assumptions made:

- Meter has been installed on date of installation of pumpset, is in working condition since and has not been reset.
- Measurements are taken on 31/01/2014.
- Instantaneous electric power is used to calculate hours of pumping and is assumed same throughout the number of days of operation.
- Apart from instantaneous power, the hrs of pumping is calculated considering pump power rating and measured efficiency, but this gave lesser values compared to shown in table below. Hence are neglected as conservative measure.

It has observed that the average annual hours of pumping came out to be 422, which is far less than considered for project. During first 3 years of project there was report of draught which may have reduced the pump usage. Also, we are not sure if our assumptions are true. hence a firm number of pumping hours cannot be concluded from this. Whereas, the maximum number obtained is 1693 hours which is more than 1640 hours, only for single beneficiary. In addition to this only 6 out of 101 measurements shown the pump usage more than 1000 hours. This raises the question on hours of pumping considered for the project as it seems very high (1640 hrs) compared to observed from analysing data.

Table 5.2 Hours of pumping details - based on metered readings of M&V

Avg. Meter reading (kWhr)	Avg. no of days	Annual hrs of pumping	
		Average	Max.
4070.35	774	422	1693

(Source: M&V measurement spreadsheet authority of MITCON & MSEDCL)

5.3.3 Scenario 3: From Survey

As a part of this investigation, during the survey conducted by me, beneficiaries were asked about their pump usage pattern, based on that an approximate pumping hours has been found out. From this farmer survey it has observed that, there hasn't been any changes in their watering

pattern/hrs of pumping with usage of new pump. And approx. annual hrs. of pumping came out to be **1439hrs** for about 15 farmers with sufficient water available for irrigation. And approx. **1136hrs** including the ones who mentioned the issue of unavailability of sufficient water (n=19).

5.3.4 Scenario 4: Metered - DPR data

Table 5.3 shows the annual hours of pumping based on metered energy consumption. From detailed project report (DPR) of scheme, the data for annual energy consumption of meter consumer categorized according to sanctioned HP is taken. This is shown in column (A). Average power rating for different HP ratings are calculated based on M&V data for old pumps, appendix 2 contains the details.

It has observed that the average pumping hours considering only these 4 types of IPS is 1154 hours. Which is again very less than 1640 hours.

Table 5.3 Hours of pumping details - Based on metered readings under DPR

Pump rating	Avg. Annual Energy consumption (kWhr) (A)	Avg. Power Rating of old pumps (kW) (B)	Annual Hrs. of pumping (A/B)
3 HP	4554	3.75	1214
5 HP	7385	6.34	1165
7.5 HP	8720	9.12	956
10 HP	13065	10.2	1281

(Source: DPR, 2009)

5.3.5 Scenario 5: Crop water requirement

Table 5.4 shows the details of analysis for hours of pumping based on cropping pattern in Mangalwedha. The results show the annual hours of pumping to be 1183.

Considerations:

The average agricultural land holding in Solapur district is 4.6 Acres, while the assumption of irrigated land per farmer taken here is more and is conservative in nature. (Census 2011)

Sugarcane and Rabbi jowar are major crops in Mangalwedha and contributes to about 75% of land. Based on area occupied by these crops in district, 20% of irrigated land is assumed to be

occupied by sugarcane and rest by Rabbi Jowar. As both of these are water guzzling crops outcome will be on conservative side.

For pilot project about 2221 pumps were surveyed, for which the average discharge of 22.16 m³/hr has been found. The same has been used to calculate the hours of pumping here.

Table 5.4 Hours of pumping based on crop water requirement

	Sugarcane	Rabbi Jowar	References
Area under irrigation per farmer (acres)	4.95		Sample survey of 10 farmers
Approx. land under crop (Acre)	0.99	3.96	
NWR for Normal rainfall (in mm)	2020.7	398.3	For Mangalwedha block
Discharge (m ³ /hr)	20.16		MITCON 2017 - Baseline survey
Field application efficiency (%)	60		FAO – for floor irrigation
Total water required (Cu.m)	13336.62	10515.12	
Hours of Pumping	662	522	
Total hours of pumping in a year	1183		

5.4 Concluding Remark:

From all five scenarios, it can be concluded that there is high chance that hours of pumping are less than 1640 hours which was proposed during DPR. All 5 scenarios have shown the annual hours of pumping to be near about 1200 hours. This is going to affect the project benefits negatively, as calculated energy savings and hence financials will be lowered. Calculating the exact hours of pumping can be a critical task, and coming to a certain value is difficult. But finding the certain range is possible and care must be taken that the approximated value lies nearby the range. (Note should be taken that these calculations are for Mangalwedha block and hours of pumping may vary from place to place).

Chapter 6

M&V ERROR ANALYSIS:

6.1 Finding the existence of error:

Figure 6.1 represents the base measurements of total head and efficiency of EEPSSs from secondary data. Following analysis is done to see if we can spot the possibility of error due to friction head measurements and/or calculations.

The selection of pumpsets were appropriately done, as per the mandate of pilot project. If this is true, then there should not be much variation between efficiencies of two pumpsets with same total head. But, from chart not only the efficiencies more than 100% has observed; which is impossible but also large number of IPS seems to have different efficiency (with significant difference) at same head.

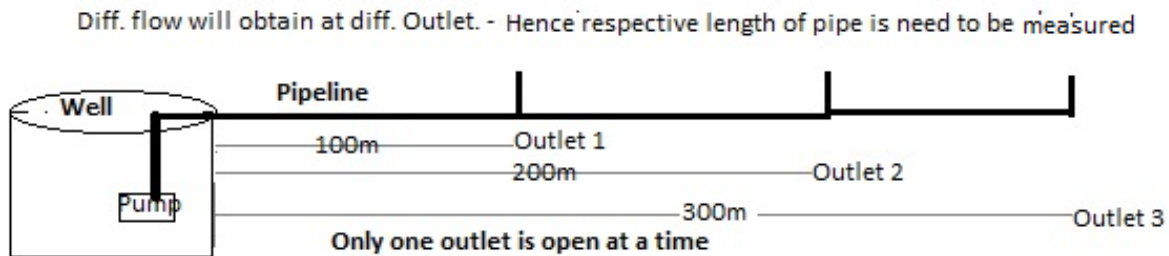


Figure 6.1 Schematic of tapping point error

Set A represents the measurements having less than 3 m of friction head, which is very small and will have a negligible effect on efficiency, making efficiency a major function of static head. While Set B represents the points with friction head more than 9 m, which can significantly affect the efficiency. Now if we compare the behaviour of Set A & B, for the points with large efficiency difference at same total head; it can be said the difference is a function of friction head. Hence if point is unreliable based on efficiency value, there is possibility that error due to flow rate measurement (pipe length and friction) exists. From figure 6.1 it is concluded that,

there is error in friction head measurement and calculation, and its presence can be found by mere data analysis.

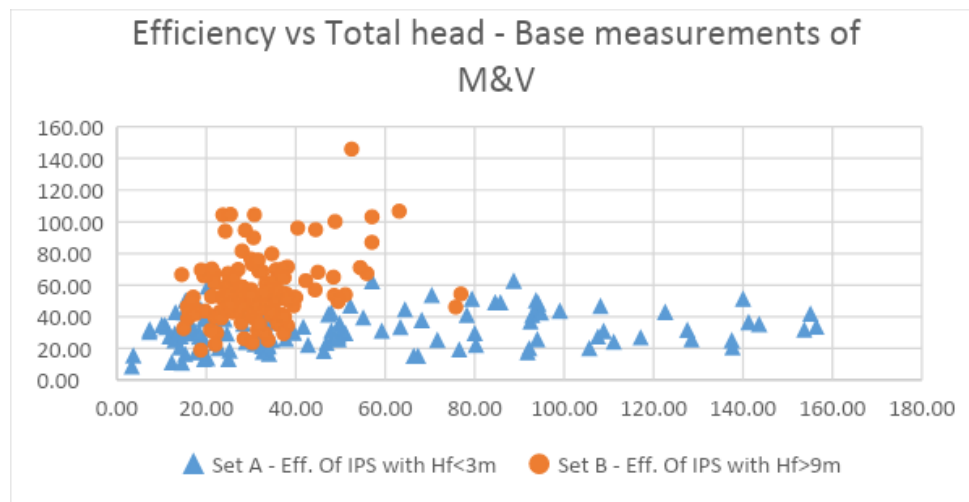


Figure 6.2 Efficiency vs Total head - Base measurements of M&V

Figure 6.2, shows study of M&V measurements for CSM 3S(5) type pumpset installed under the pilot project. As we can see the measurements shown in circles are out of pump curve and practically not feasible to achieve. This proves the existence of error in measurements. Based on the empirical observation, the reason behind this can be usage of different pumpset by farmer than the one that was installed in the scheme. As it is submersible it is difficult to know about variant of pump during measurements. Then calculation error and flow rate measurement error are another reasons.

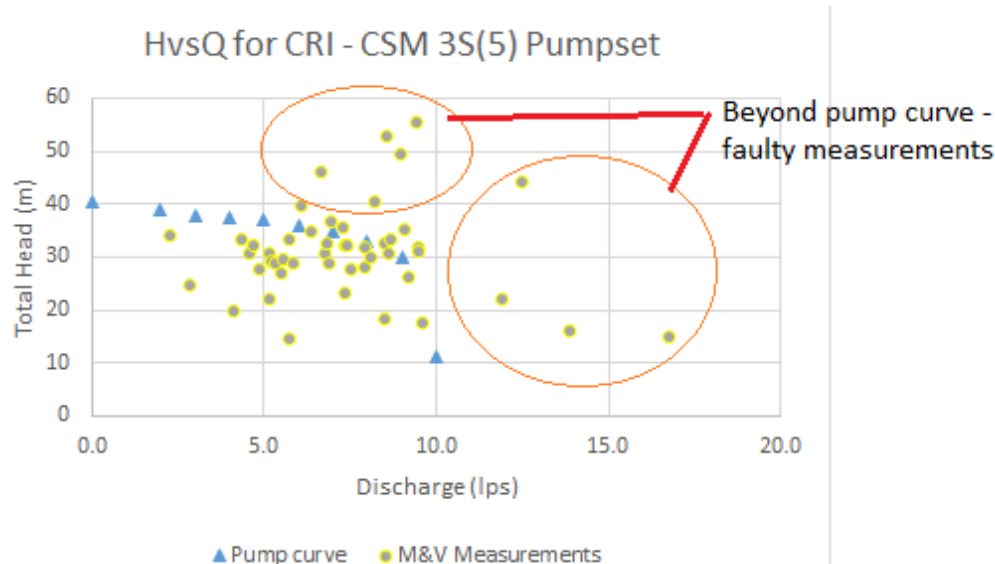


Figure 6.3 Finding error through single case

6.2 Investigating the extent of error:

6.2.1 Description:

Efficiency calculation and measurements error analysed are shown in table 6.1 From Primary analysis; the existence of few had come to our attention. Hence, field level measurements were taken during the 5th year M&V process of the pilot project. M&V agency has prepared a form for noting down the measurements, can be referred through appendix MITCON form. These measurements then were used by officials in M&V agency for efficiency calculation and detail energy saving analysis. These measurements noted down in form during M&V process is shown in columns under Set A. While column B represents the corrected efficiencies methodology for which can be referred from Appendix 1. Where as in Set C, the efficiency is calculated with the methodology similar to that was used by M&V agency. These calculation methods are almost similar except that; M&V agency has used the constant value of friction factor. In their methodology major difference is the value of friction factor which is 0.006 and considered same through. While the friction factor in corrected value is calculated for each reading and is near have average value of 0.017.

From field observations, length of tapping point and elevation head found to be a concern for error, as no proper equipment were used for these measurements. Tapping point is the location on pipeline where flow rate is measured i.e., discharge is tapped. While, the depth of water surface from ground, which contributed majorly to static head has been measured with plumb bob for dug well, and has been considered as correct for comparing all the calculations. Borewell depth could not be measured, it was asked to the farmers, and is considered same and correct for comparing calculation as our focus was on to check error due to tapping point variation and elevation measurement.

6.2.2 Analysis:

Following Nine readings shows an error from negative to positive. The details of these nine readings are available in appendix 3. The error is calculated with respect to Set B, i.e., primary data. In Sr. No. 1 (reading 1) the error between set A and B is maximum which is about 133%. The reason for this being the huge difference between considered pipe length. While at few points difference between elevation noted by M&V personal and by author, resulted into balancing the error. The actual flow measurements were done at the pipe length (tapping point) of 153 m. While the pipe length noted down in M&V measurement form was 670 m. Which was total pipe length installed in a field in most of the cases. This resulted into usage of higher into calculation than actual one.

In 8 out of 9 sample points, there exists the tapping point error. It can be said that if same measurements and calculation methodology is used, then results will show the average error of about -36.87%, a very high error. While from Set C it seems that the lower value of friction factor has accounted as compensation for these errors, as the average error is 13.22%.

Table 6.1 Error in M&V Measurement & Calculation

Sr. No.	Set A - M&V measurements only (Secondary data)	Set B - Our's/ corrected (Primary data)	% error (100*(B - A)/B)	Set C - With M&V calculation and measurements
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	Lengt h (m)	Slop e (m)	Efficienc y %	Lengt h (m)	Slop e (m)	Efficienc y %		Efficienc y %	% error (100*(B -C)/B)
1	670	0	60.98	153	3	26.15	-133	30.76	-17.63
2	396	1.22	73.16	91	3	45.93	-59	47.85	-4.18
3	300	0	48.74	168	1	33.83	-44	25.57	24.42
4	326	2.7	42.18	71	1.6	21.48	-96	26.39	-22.86
5	447	2.7	28.96	140	-0.5	20.93	-38	12.83	38.70
6	58	0.6	40.29	40	1.5	38.99	-3	27.16	30.34
7	412	1.2	25.04	181	2.6	24.83	-1	22.53	9.26
8	386	2	16.28	412	9	23.15	30	13.57	41.38
9	406	3	16.66	406	6.5	19.37	14	15.59	19.51
			39.14			28.30	-36.87	24.69	13.22
			18.46			8.69		10.19	

6.3 Case Study - Sensitivity Analysis for error:

From section above it is clear that there exists a substantial amount of tapping point error. As farmers have large fields and hence the multiple tapping points, their operating efficiency will be different with each point. Hence during measurements care need to take about tapping point for flow. To see the quantitative extent of such error following sensitivity analysis is performed.

Table 6.2 shows the sensitivity analysis with the help of two examples, based on primary data collected. In both the example, first reading is real measurement, while rest three calculations are done to show error that could have occurred; if tapping point information (length of pipe) is not collected properly and same measurements are applied to other tapping points. The real

measurements of input power and discharge are assumed constant, as this is the error performed during M&V process. There were two delivery pipes, length of first pipe being very small; producing very small error, the change is neglected. Invariably, pipe two has multiple chambers, and incremental change of 100 ± 5 m is assumed as different tapping points, till the actual maximum length of pipe of field is achieved.

In example 1, measurements are done very near to pumpset i.e., at 61 m of pipe 2 length. Hence the actual discharge will be more than at extreme end, but as discharge is considered constant it will produce higher frictional resistance with larger lengths. While in example 2, measurement is taken at extreme end i.e., at 392 m of pipe 2 length. Similarly, the discharge at extreme will be lowest which will result in lower frictional resistance value for smaller lengths as actual flow would have been more.

Example 2, shows negative incremental error with average of -4.54%, suggesting calculated efficiency will be lower, producing false data of energy savings i.e. lower than actual. Similarly, Example shows a very high positive incremental error with average of 17.88%. This will produce false results with higher energy savings data than actual.

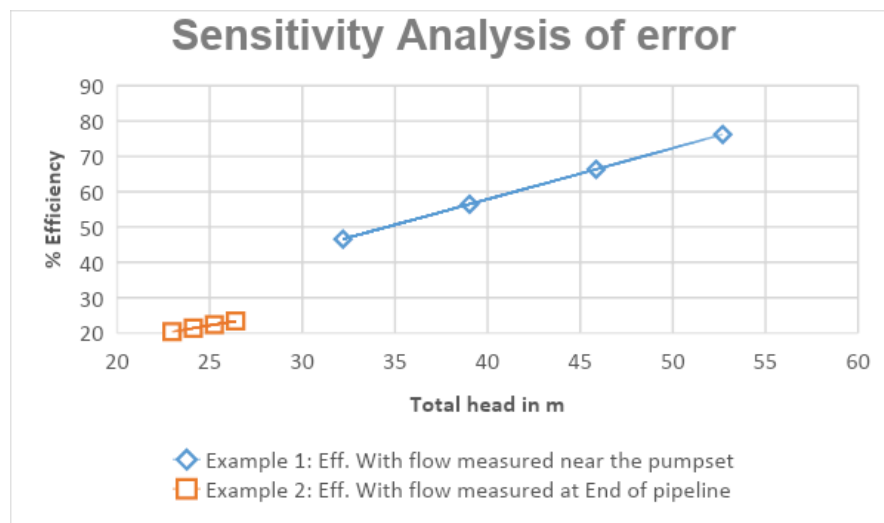


Figure 6.4 Sensitivity analysis of Error – with every 100m change in pipe length

Table 6.2 Error due to mistake in tapping point variation

	Efficiency %	% incremental error in Eff.

Example 1: Static head = 26m Flow rate = 6.99 lps Input Power= 4.74 kW	46.57	
	56.45	21.23
	66.34	17.51
	76.23	14.90
Example 2: Static head = 21.5m Flow rate = 4.26 lps Input Power= 4.72 kW	23.37	
	22.36	-4.33
	21.35	-4.53
	20.33	-4.75

(For details refer Appendix 3)

Concluding Remark: It's better to take measurements at some midpoint which represents an average of the usage efficiency. Also, the point needs to be specified in the measurement protocol so that the same point is selected for every test as error produced will be lower, and will be on conservative side. Out of 430 farmers 301 has 2 or more chambers (which can be possible tapping points for flow measurement). This shows the significance of tapping point in the measurements.

Chapter 7

EXTERNAL FACTORS AFFECTING IPS EFFICIENCY

7.1 Significance:

Field conditions to pumpset usage varies widely. Hence, how various factors affect the pump performance may vary in field and pumpset testing laboratories. In addition to this, manufacturers do not share the information related to performance variation in pumpset. Under M&V of pump replacement project field level measurement of one of such performance measure – efficiency has been taken. Such dataset, involving actual end user scenario can be used to increase our understanding of IPS. This can help in taking improvement measures in regard to scheme and pumpset performance. For example, the objective of scheme was to achieve energy saving with improved efficacy of IPS. With prolong usage of pumpset there is possibility of efficiency drop due to deterioration of components, as it has many moving parts. Now, with some deterioration rate after some period there will be a condition when energy saving won't be

achieved. Considering this the financials of scheme, precautionary measure to reduce the deterioration etc. can be planned for future projects.

7.2 Time & Breakdown Vs efficiency: Case Study CSM 3S(5)

Under this scheme 30+ type of pumpset models were replaced. Performance behaviour of each is different. Openwell 5HP pumpsets are widely used by the farmers in Mangalwedha area. As it is more accurate to measure head in dug well over borewell, a single pumpset model of category (CRI CSM-3S(5)) is selected from the available data for analysis. A single instantaneous measurement for all pumpsets were taken, and we did not have multiple reading for the pumpsets. But with different users with different head and discharge conditions were assumed as different pumpset operating conditions.

For the analysis, all the measurements for selected pumpset model has been extracted from available secondary data. This mainly included beneficiary's measurements taken during 1st and 2nd M&V process of scheme. Then this data has been processed to see the effect of various factors like voltage variation, aging and breakdown-repair on efficiency of pumpset. Low voltage seemed to be a problem for farmers, while aging and breakdown-repair has an important place in scheme; as for 5 years of project duration free repair was provided, hence focus has been given on these factors. The detail methodology and results are discussed below.

7.2.1 Methodology:

Figure 7.1 is pump characteristics for pump CSM-3S(5). The pump curve is prepared from head and discharge data obtained from manufacturer's pump catalogue. While each M&V measurements represents the instantaneous measurement done by M&V agency for different beneficiaries. As we have already discussed in section 6.2 about possible errors in M&V measurements, following methodology is used to reduce those errors.

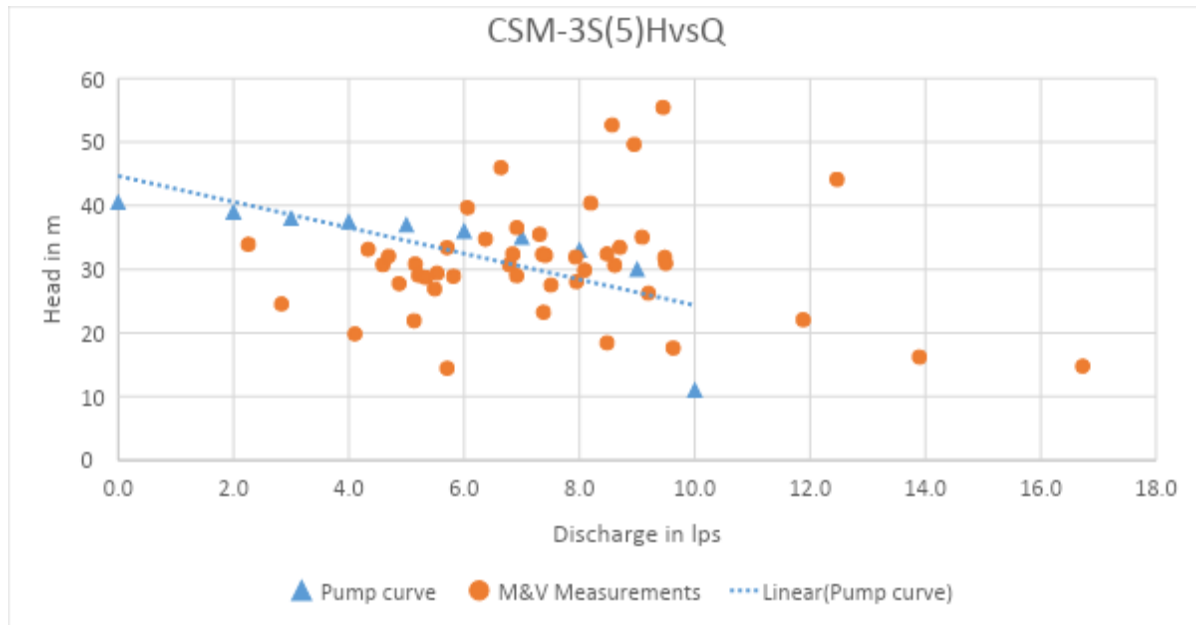


Figure 7.1 Pump Curve for CSM-3S(5) pumpset

- From field observation during M&V process, it has been realised that the electrical input and discharge measurements can be trusted.
- Hence from pump curve a pump curve equation is obtained as shown below.
- Equation for pump curve: $H = (-0.1192) Q^3 + (1.3688) Q^2 - (4.4984) Q + 40$
Where, H is Theoretical head, at Q i.e., measured discharge.
- And for given discharge value, the theoretical head is found out.
- Based on nature of these value any ambiguous data has been removed. e.g., negative head with measured positive static head is not possible, data points which are too deviated from expected behaviour i.e. pump characteristics, points having efficiency more than 90%; which is not possible for this pump & at low voltages.
- Now based on this new theoretical head and measured discharge, theoretical hydraulic output power is calculated by the methodology discussed in Appendix 1
- From this theoretical output power and measured electrical input power, theoretical efficiency has been calculated.

7.2.2 Effect of breakdown-repair on pump efficiency:

From the shortlisted 45 pumpsets, those which has been broke down and repaired before the M&V measurement were found out from list of broke down pumps. Eight of such pumpsets out of 45 has been found. Following figures, A&B shows the average efficiency of these broken down - repaired pumpsets and those which did not faced failure before M&V. Figure A shows the average efficiencies measured and recorded by M&V agency. While Figure B shows the average of theoretically calculated efficiencies of pumpsets, as mentioned in above section. Instead of unreliability of M&V data, its average efficiency is considered here to see if we get the same pattern as theoretical calculations. Same pattern suggests, though with some error; M&V agency if had done such analysis would have gotten the similar results.

Figure A&B shows the average efficiency of set of pumpsets at base i.e., at the time of installation; when it was new and at M&V process. The important thing that that we need to consider here is, this variation is not a function of breakdown-repair alone. As different pumps were installed at different time and the measurements were also taken at different times, and voltages, operating head and other field conditions, effects of all these parameters is likely to be there. Hence, the quantitative difference in variation cannot be justified at this stage. Although the trend in change can be visualized easily. Also, it should be taken into account that separate study of such factors is difficult on the field.

From graphs we can see that the efficiency of pumps which has failed and repaired (At M&V) has gone up compared base value. To confirm if breakdown has affected this change or not; a comparison of pumpsets which never broke down is made. And it seems that their performance has gone down compared to base value. The reason for this variation is not clear and need to be investigated further. Though the change of pumpset components like winding, impeller may have affected it as these are the changes takes place under repair.

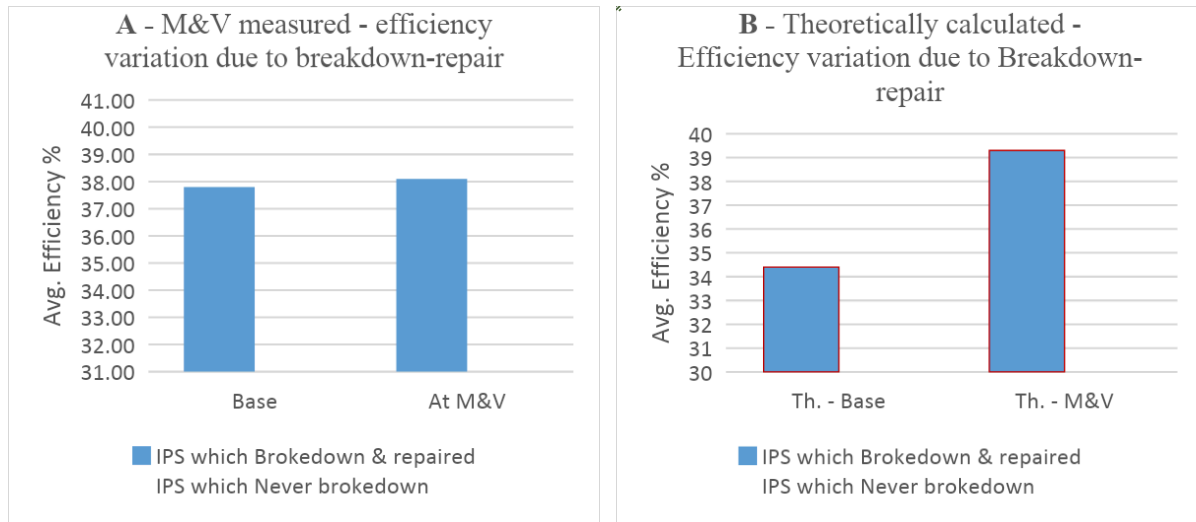


Figure 7.2 Effect of breakdown on efficiency of Pumpset

7.2.3 Concluding Remark: It has observed that efficiency of IPS has increased after repair. But, only 8 pumps were brokedown and concluding based on this won't be fair. Though this methodology can be used as primary study for further investigation.

7.2.3 Change in Efficiency with time:

There is not any formal study of efficiency decline over the time, TERI examined electric pumpsets several months after rectification and found an average fall in efficiency of approximatly 5%.(Reidhead, W., 2001) To see how the efficiency of IPS varies with time in field following analysis is performed. From this pumps which has brokendown-repaired has been omitted, also the voltages of sample pumpsets varies found to be within operating range. Following is the methodology used for analysis.

- Difference between installation date and M&V measurement date has been calculated and is termed as approximate number of days of operation.
- Theoretical efficiencies were foundout as discussed previously, and the averages of these for sample IPS has taken.
- Samples are number of IPS used for this perticular analysis. Each set is selcted based on number of days of operation/age. The sets were selected such that, the deviation from average number of days considered is least to include maximum number of IPS in sample set.

- From each of these sets, their average efficiencies are taken.
- Table 7.1 shows the results of analysis done.

Table 7.1 Efficiency variation with age of pumpset

	Mean measured(M&V) efficiency - variation with time				Mean theoretical efficiency - variation with time			
Age of IPS (Days)	406	787	1345	Mea n	406	787	1345	Mea n
No. of IPS used for data	5	5	3		5	5	3	
Avg. Base measurements	28.12	39.86	46.67	38.22	37.3	35	45.6	39.3
At M&V/ After - days	26.48	35.13	38.73		34.8	29.4	35.6	
Drop in efficiency per year	1.47	2.19	2.15	1.94	2.25	2.60	2.71	2.52
% drop in efficiency in a year	5.24	5.51	4.62	5.12	6.03	7.43	5.95	6.47

Date of M&V measurements itself has assumed constant, which may vary by ± 30 days, inducing minor error.

These results does not show the variation in efficiency solo due to time, as various external factors over the time may have affected the performance differently. But, it helps us to understand the change in performance of IPS in field over the period considering all the external factors.

Figure 7.3 graphically represents the change in efficiency for different IPS sample having different time period of operation/age. It is clearly seen that there is drop in efficiency as IPS gets older. Also, the difference between efficiency seems clear with increase in age, in most of the cases. From theoretical calculations it seems that the efficiency of IPS is dropping on an avererage by 2.52 each year which is about 6.47% of base measurement every year.

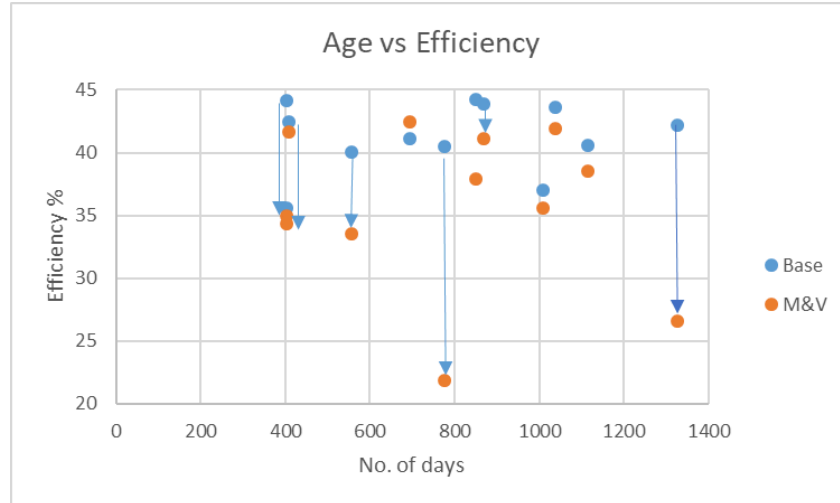


Figure 7.3 Variation in efficiency with age of pumpset

There are various reasons causing this performance deterioration over the time, one of them being wear and tear of moving parts in pumpsets. Two root causes resulting change in IPS physically are, deformations of pumpset components due to friction and temperature increase; which can be cause by frication as well. Hence efficiency variation with aging factor is need to be considered for improving the pumping performance in the fields. Also, for this perticular ESCO project, not only improved efficient pumpsets were required to achieve the desired objective. But also needed to sustain the eefficiency with time/age. Knowing the efficiency deterioration rate can help in better planning of such projects.

Cloncluding Remark: Under the agreement of Mangalwedha pump replacement project, the targeted average efficiency of all IPS was 39.5% as base, which was expected to be 34% by the end of 5 years. (MITCON, 2017) That is the deterioration rate of 2.5% of base value per year. But here we have found it to be 6.49%. This suggets the results from the field can be different and need to be considered while planning.

7.3 Effect of variation in voltage – Experiment

7.3.1 Introduction of Experiment

Through the ESCO pump replacement project in Mangalwedha old local pumps were replaced with 5star rated energy efficient pumps. Except that EEPS has higher efficiency than local pump we did not know anything about how different factors affect energy efficiency. In addition to this, from literature study and field observations we have found that the voltage fluctuation is a big issue for farmers.

As EEPS are costly compared to local make pumps, from financial point of view keeping the objective of energy saving constant, we thought what if we replace old local pumps with only 1star rated or local make pump and provide the maintenance. But to know this we had to know the proper differentiation between local make pump and EEPS.

Hence an experiment is conducted to compare these two pumps for efficiency variation. As it would have been difficult to build the experimental setup due to time and financial constraints, we selected a pump repair shop in Sangli district of Maharashtra, which had a proper pump testing setup. Also, purchasing pumps from local market (region in which agriculture is prominent) gave us the advantage of selecting pumps which are commonly used in that locality.

The selection of pumps for experiment was done based on our knowledge and experience on field. We selected two submersible pumps one with BEE energy 5-star rating with ISI mark and another with non ISI mark. The care was taken, that the pump must be similar in terms of operating point (based on specifications mentioned by manufacturer). The details of both the pumps are mentioned in table 7.2.

Table 7.2 Details of Pumpsets used in Experiment

Manufacturer	Waterman	Laxmi
model	SHM 54 A	BV95252
Category	5star ISI mark	Non ISI mark
Pump rating (hp)	5	5

7.3.2 Design of setup:

A setup at pump repair shop included a (1 X 1 X 3) m³ water tank in which pumps was submerged with the help of chain and pulley.

Power controller unit, from which voltage could be varied by certain steps. 3 phase stabilize was present, hence the voltage at all phases were maintained same. The voltmeter and ammeter readings were crosschecked without own power analyser, and no deviation was found between two.

A mounting over the pump outlet, which consisted of pipe with throttle valve and arrangement to join pressure (head) gauge.

A pressure gauge was used to see and maintain the total pressure generated by pump.

A 250 litre container was used for the purpose of measuring the discharge. And a pipe which was manually held at mounting (discharge point) over pump to allow the flow into the container. For each reading (variation in head and/or voltage) container was filled and time to fill it was measured with the help of stop watch. After each reading container was emptied into the same water tank from which water was pumped. Pumps were tested one after another for same set of change in parameters to study the difference between them.

The setup served the purpose to study and analyse the pump characteristics at primary level.



Figure 7.4 Experimental setup

7.3.3 Analysis:

H-Q Curve Comparison

Figure 7.5 pump curves for both the pumps and pump curve for waterman pumpset given in catalogue. These curves are drawn for measurements taken at rate voltage of 412 ± 5 volt. It is observed that, as curves almost coincides; there is very less error between manufacturer's and experimental values. The slight error exists due to flow measurement error because of leakages at pipe mountings. In addition to that we can conclude that for same head Laxmi (non ISI mark) pump gives the higher discharge. But, the input power consumption of Laxmi has to be taken into consideration as well.

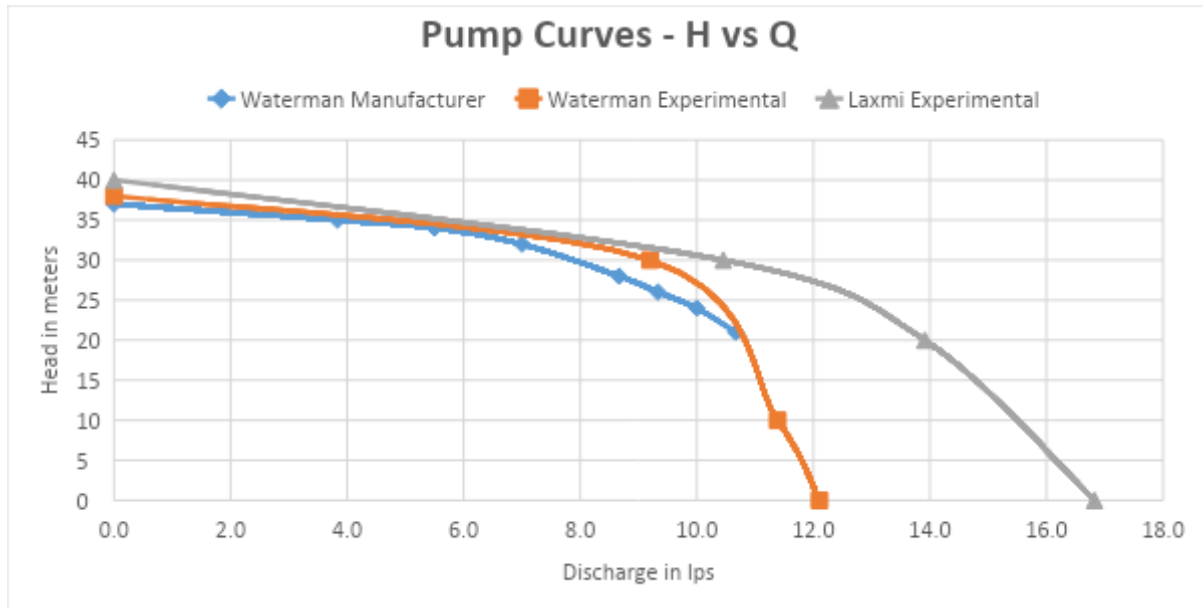


Figure 7.5 Pump curve (H vs Q) comparison

Power Variation Comparison:

Laxmi pumpset consumes more power for same hydraulic output compared to waterman (standard) pumpset. Figure 7.6 shows the performance of pumpsets at low and rated line voltages. At low voltage the operating range of Laxmi (non ISI mark) pumps seems to be very low compared to Waterman (EEPS) pumpset.

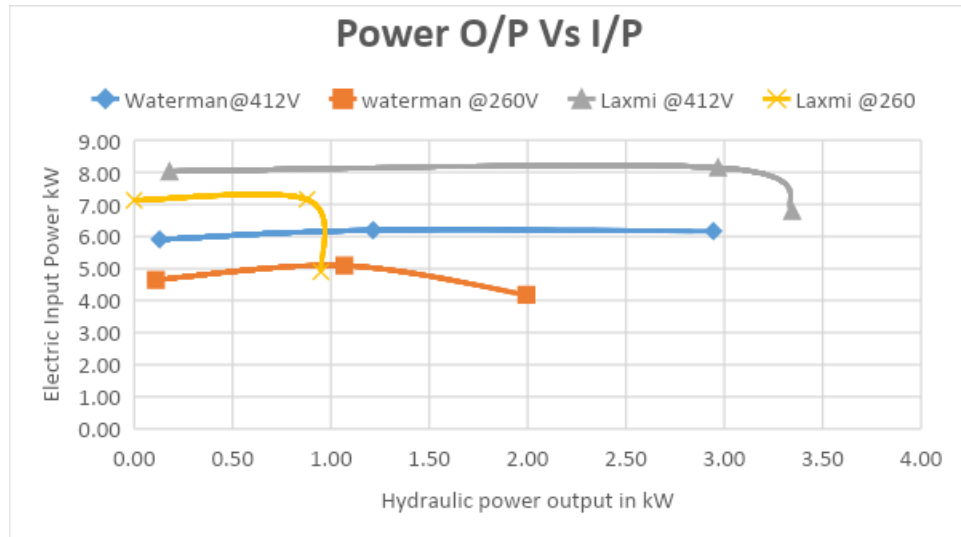


Figure 7.6 Variation in output of pumpset with input power variation

7.6.4 Variation in efficiencies with voltage:

Figure 7.7 & 7.8 shows the efficiency vs discharge graphs for non-ISI and ISI pumpsets respectively. The difference is clearly visible. The one important thing to note here is with decrease in voltage the performance of pumpsets deteriorates continuously. With non ISI mark (nonstandard) Laxmi pumpset this deterioration seems more compared to EEPS Waterman. Till Waterman pump is in operation range of voltage the deterioration is slow compared to when it operated at low voltage levels. From this we can conclude that EEPS waterman is better choice when pumpset is need to operate at varying voltage and loads.

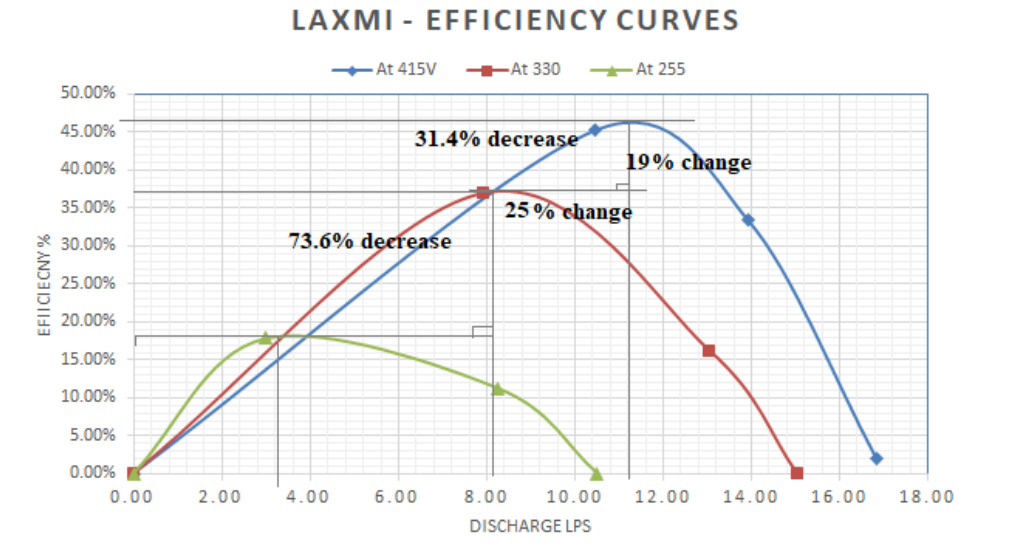


Figure 7.7 Effect of volatge on efficiency for Non Standardized Pump

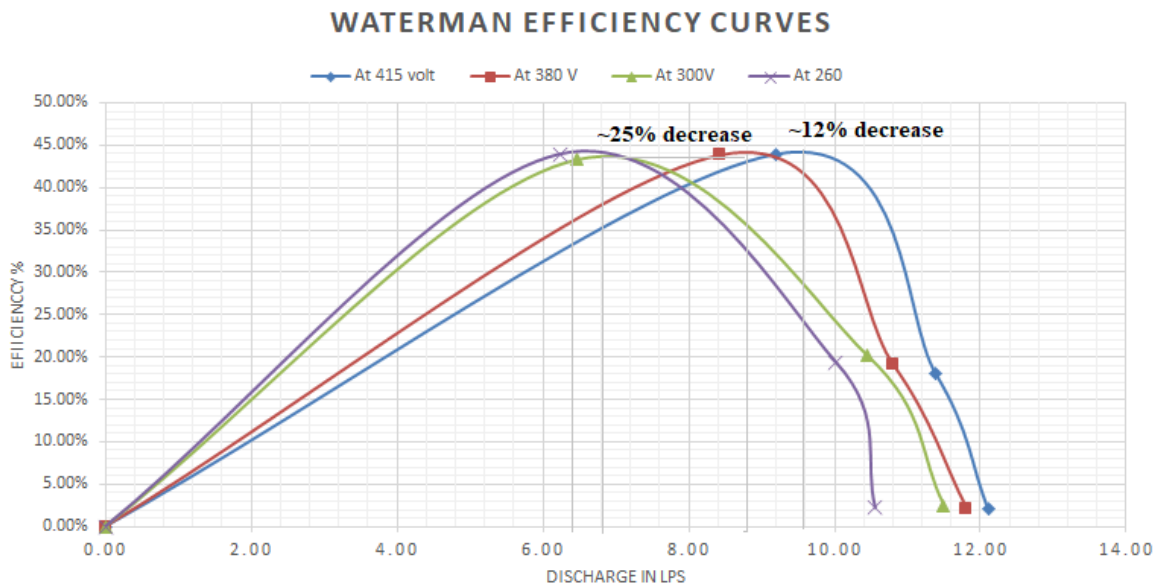


Figure 7.8 Effect of volatge on efficiency for Standardized Pump

Figure 7.9 shows the performance of both the pumps for different heads and discharge at different voltages. Though with variation in voltage curve shifts to lower operating points the overall pattern/equation of curve remains same for both the pumps. When done the comparison the downward shift occurred in Laxmi pumpset is observed to be more compared to Waterman pumpset. This implies the importance of standards, as standard pumpset is able to perform still better than nonstandard one. If we observe the slopes of each curves the slopes are pretty stiff for Laxmi pumpset compared to that of Waterman's for certain range of load. While at higher loads Waterman's pump curve become too stiff compared to Laxmi pumps. The significance of this difference is that, in fields voltage and head varies pumpset needs to operate at different conditions; which requires pumpsets tolerating the sensitiveness in performance keeping it optimized for maximum amount of conditions.

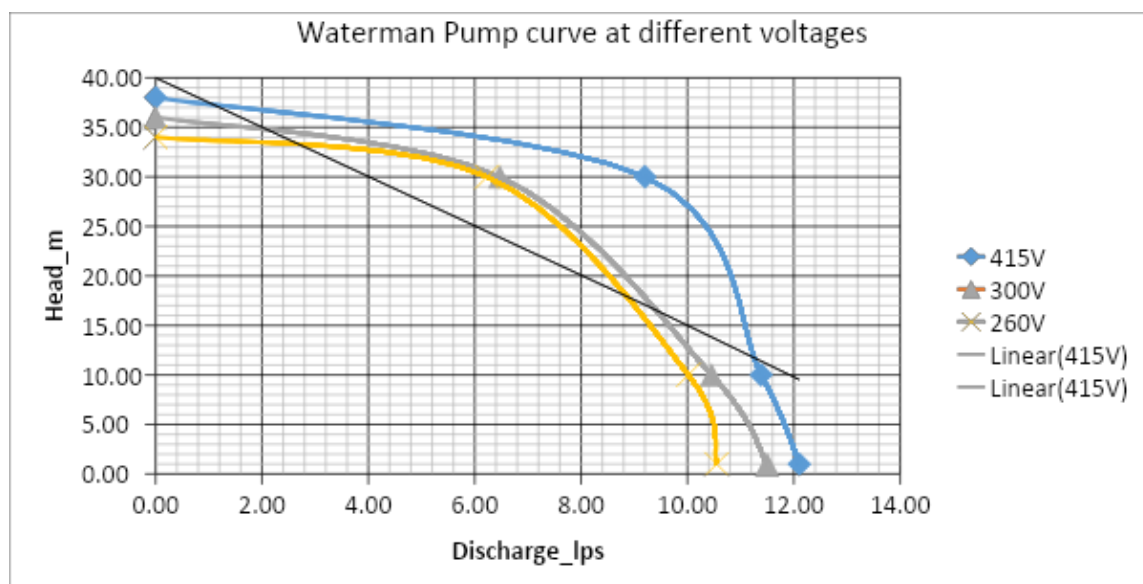


Figure 7.9 Effect of volatge on pump charecteristics for Standardized Pump

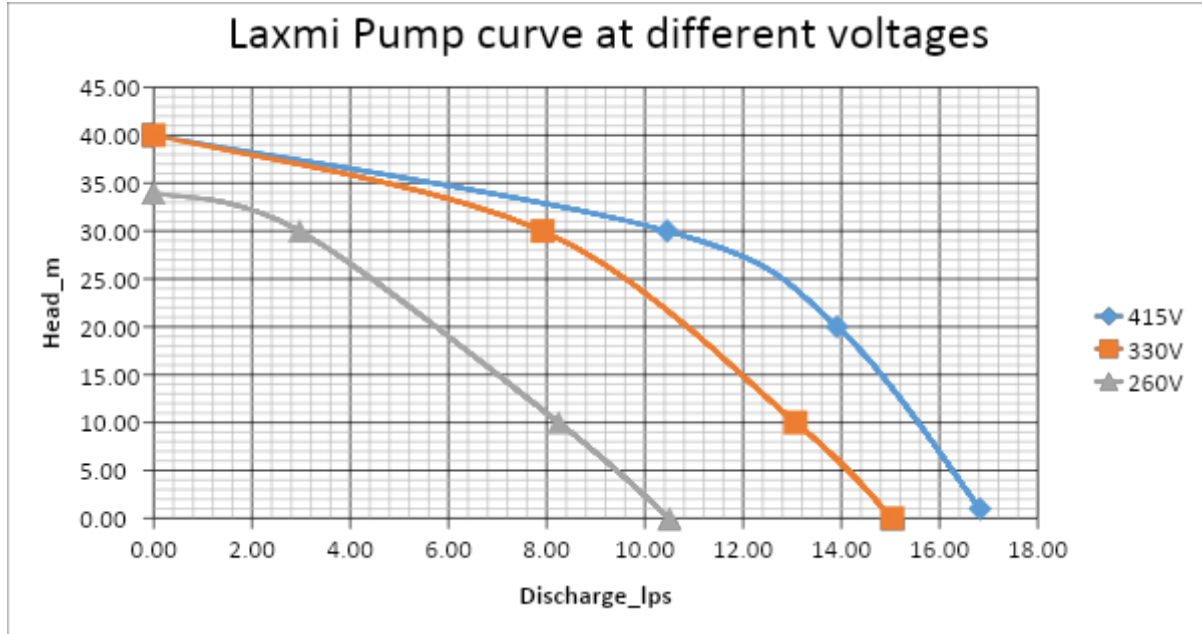


Figure 7.10 Effect of volatge on pump charecteristics for Non Standardized Pump

From efficiency curve at rated voltage at has observed that maximum efficiency is observed at head of about 30 meters for both the pumps. As head was the controlling parameter in experiment, it will be easy to observe performance variation keeping head constant and same for both pumpsets. Hence to see the variation in performance of pumpsets, specifically efficiency with voltage head is kept constant. Figure 7.11 shows the plot of efficiency versus voltage for both the pumpsets. As pumps are new, operated at operating head, negligible head variation the deviation in performance can said to be majorly due to voltage change and not affected by inappropriate pump selection, aging, breakdown and head. As we can see there is efficiency of pumpsets on y – axis and voltages ate which measurements on taken on x-axis.

Equation ($Y_1 = 1E-05X_1 + 0.4338$) is for the trend line for waterman pumpset representing linear relationship between voltage and efficiency. While equation ($Y_2 = 0.0023X_2 - 0.4638$) represents the trend line for Laxmi pumpset. The slope of trend line for Laxmi pump is far greater than that of waterman pump ($0.0023 > 10^{-5}$). That is, small variation in voltage causes large change in efficiency for Laxmi pump and very small for Waterman pumpset. From this we can conclude

that the performance of Waterman (Standard) pumpset is better than Laxmi Pumpset at lower or fluctuating voltages.

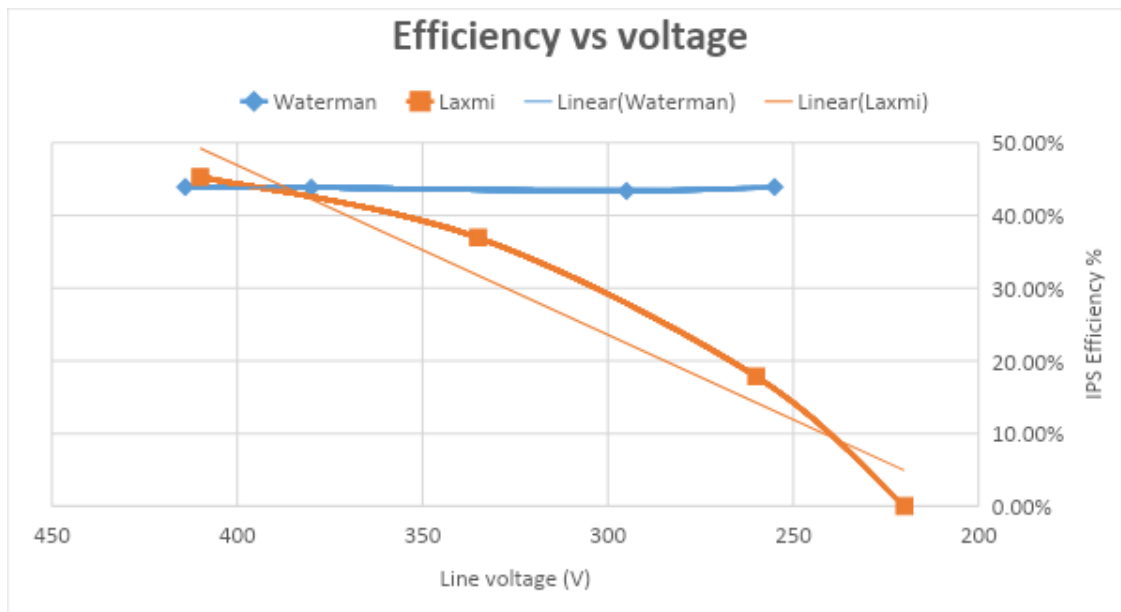


Figure 7.11 Comparing ISI & Non ISI pumpset for efficiency variation with voltage

Chapter 8 LEARNINGS & DISCUSSION

8.1 Interaction with DISCOM:

Following are the personal opinions of officials from utilities directly or indirectly associated with the pilot project:

- Scheme has benefited to utility financially, and can be expanded on similar basis.
- Farmer awareness and education about the electricity usage, considering the theft, bill payments etc., are need to be focused upon to improve the efficacy of utilities.
- Theft of electricity is large issue, unavailability of enough taskforce makes it difficult to tackle this. As every wireman has allotted about 3-4 villages managing it by themselves becomes difficult. Technology intervention is needed to stop the theft.
- Absence of strong action against bill defaulters and backup from political agendas is one of the major reason for low collection efficiency.
- Farmers are billed based on sectioned HP rating by flat rate tariff. To reduce the bill amount, farmers get sanction of lower HP than actually installed resulting the mismanagement of load.

8.2 Interaction with Pumpset Manufacturer:

Pump manufacturer has direct control over the performance of pumpset, except for external factors. In this pilot project, implementation agency “CRI pumps” was a manufacturer as well. Hence to get manufacturer’s perspective a visit has given two pump manufacturers in Rajkot, Gujrat - a major pump manufacturing hub in India.

Following figure 8.1 Represents the distribution of IPS of different categories, in India. It is seen that the BEE star rated pump distribution is very less in India, which were used in pilot project. While ISI mark pump are easily available in the market. It has observed that after period of operation efficiency of EEPS deteriorates and come close to efficiency of pumps with ISI marks, also with varying operating heads on field efficiency fluctuated. EEPS being costlier than just ISI mark pumpsets, increases the cost of project. Hence we wanted to know what if usage of ISI mark pumps is promoted and maintained properly. Keeping this in mind few points were discussed with manufacturers following are the few inferences drawn from discussions.

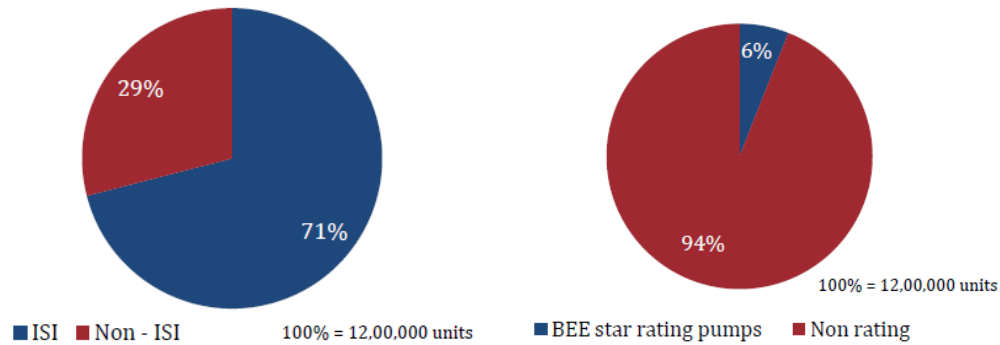


Figure 8.1 Scenario of standardized pumpsets in India

- It costs only about Rs. 10-20 per pumpset to get the ISI mark (In bulk).
- But, to get this mark a proper pumpset testing facilities are required and it is expected that pumps of same model adheres to claimed performance. Many manufactures used inferior quality material (low material rejection compromising quality) and lacks a proper testing facility, this is one of major reason for not getting ISI mark instead of being affordable.
- ISI mark or BEE star rating are given based onetime testing in labs, it does not account for performance deterioration with prolong usage.
- Hence, only usage of ISI mark pumpset instead of EEPs; is not going to guarantee the savings in budget as performance of ISI standard pump may deteriorated further.
- ISI mark for efficiency is provided based on observations at operating point (BEP) and there is no firm explanation about allowable efficiency variation at with variation in characteristics.
- Pumpset material is an important aspect affecting pump efficiency, as wear and tear with prolong usage can deteriorate efficiency. There is no provision in ISI standard for efficiency (IS8034:2002, IS 9079:2002, IS14220:1994) for material restriction. Though it is good, as it gives freedom to manufacturer to use material of their choice, amendments are needed to accommodate the any discrepancies due to material usage, in maintaining efficiency.

- In opinion to these manufacturers, the specified standards by BIS itself are too low, making. These norms for standards are need to be revised.
- If manufacturers achieve efficiency more than maximum specified standards, there are no market incentives. For example, a pumpset which will be have efficiency of 6 star will still be considered as 5 star, as BEE rating is limited to it.
- Farmers do not care much about standards, in such scenario there are no incentives for manufacturers to promote EEPs or standardised pumpsets.
- Apart from IPS, one interesting fact that brought to our attention is there are no BIS standards for domestic pumps which by volume can have significant impact.

As a concluding remark, there is need to revise the BIS standards for range of operating points, considering the material or deterioration with usage. Pump replacement project may create the awareness about EEPs / standardised IPS which will be an incentive in itself for manufacturers.

8.3 Precautions & Difficulties during efficiency measurements:

Measurements of pumpset input and output for the efficiency calculation is a most important thing for monitoring pump performance. But, measurements on fields for multiple times is not as easy as it is in laboratories. As configuration of pumping system varies from each other different issues are observed, as an obstruction to measurements. Following are few such observations made during data collection / measurements. As there is no documentation of such general things; which again cannot be generalized, these few points may help to take some precautions to take measurements.

8.3.1 Electrical measurements:

- Power supply availability and its reliability is a big issue while measuring the input electrical power to pumping system. In Mangalwedha block (where the data is collected) the power supply was available for 8 hours a day; 4 days a week in daytime, hence there was limitation to when and where the measurements should be taken. Also, transformer burnout, maintenance, and untimely power are few such issues due to which we had to wait or postpone the measurement till issue resolves.

- At few locations, particularly where the electrical panel was maintained properly; there was no opening or bare connection to connect the Power analyser probes. Hence it will be recommended to have screw driver of some sort as handy.
- For accurate measurements it's better to measure the flow rate at and electrical input at same time. Sometimes, electrical panel and the pump discharge were located at large distance for large farms etc., in such scenario having two teams with proper coordination between them may help.
- Apart from this locating the pump / discharge point and respective panel without the owner/operator is not recommended. Especially when panel and discharge point are far apart. It plays vital role as there can be multiple electrical panels located nearby (for example at bank of river, where the pumps are installed or for common well common pole may have been allotted). Similarly, common pipeline may exist and it may intersect different farm lands. Hence this confusion should be avoided.

8.3.2 Discharge Measurements:

- Measuring flowrate is a critical task, though there exist sophisticated instruments like ultrasonic flowmeter they are costly, need a skill full handling and installing it for every measurement with time constraint (power supply, farmer's time and number of measurements to be done, from point of view of M&V agency) it becomes difficult to handle. Moreover, it requires some data to be fed into like pipe thickness, material, lining etc., though not difficult, the variation of pipes on field make it time consuming, and if not installed properly accuracy will get affected. Still, for accurate measurement usage of such instrument is recommended provided the field conditions are already known and enough time is available.
- I have used the volume and time measurement (drum and stopwatch) to calculate flow rate for most of the data points. In this we need to take care of human error (while handling stopwatch), large enough container (larger the better). Finding a good container on field and a flexible pipe connection is difficult, it is recommended to carry those. from output of water chamber, water flows vertically upward which makes it difficult to collect in container, hence a suitable piping arrangement is required.

- For few measurements I calculated the flow by measuring the water column height outside the chamber, but only for large heights ($H > D$) the better accuracy can be obtained.

8.3.3 Head Measurements:

- To calculate the dynamic head, we need to know the piping configuration in details like, number and types of components attached, diameter, length of pipe, material of pipe. As for many farmers the pipeline was underground finding an exact configuration went difficult, and every farmer may not know about it as well. Hence we closed all other outlets and measured the flow through single one. Length was measured with GPS from pump to discharge point.
- Similarly, measuring elevation head altimeter/GPS is required.
- Measuring the static head for bore well was not possible, we had to rely on water depth known to the farmer (while digging the bore people/owner pays attention to depth of bore and water level, so most of the time we can assume they are aware of it. But, it at it changes it won't be accurate, an approximation can be done).
- Measuring the depth of well can be done by plump bob, precaution that need to take is, it should be of sufficient length.
- For measurement of depth of source like lake or river, usage of plump bob and GPS both may be necessary, as its depth varies as we go inside. Depending upon pump/suction location depth can be taken.
- In addition to this, the water level in source may happen to be too low to run the pump, and to take discharge measurements some stability in flow is recommended. As stable flow was not obtained for a system, due to water unavailability we had to skip the measurement.

8.3.4 Miscellaneous:

- Accompany of local personal, plays a very important role while collecting field data. Presence of farmer during measurement on their field makes data collection easy, their time and opinion are of immense importance, and accompany of local person can help them open up, makes work valuable.

- Also, finding a local contact as well as farmer (person of interest) is difficult, it is recommended to plan such things (through call) before approach.
- Travelling to these farms and villages, (where exactly to go and how) is one more difficulty anyone can face, as no public transport will go after some point in village. A two wheeler can be helpful in such case, arranging it in location unknown to us will be difficult, hence it is recommended to make pre arrangements for such travels.
- Specification plate on pump can be useful for detailed analysis (theoretical analysis) for that pump. But, getting this is impossible most of the time, in case of submersible pump checking the pump model or specs. is not possible as pumps lies in the water. For monoblock we may find the plate, but it may be corroded or missing. Asking a farmer what he bought and contacting dealer may help.

8.4 Inputs/Suggestions:

There is huge opportunity for energy saving but the potential saving of each system is a function of the inefficiencies within that system. (Bredin, M. S.,2017). There is no generalised methodology to improve the efficiency of pumping systems. As we have seen, there are various options for improving pump efficiency just from selection of appropriate pump to replacing the complete system with more efficient one. As Some systems will be efficient and only certain aspects of others will be inefficient. The trick is to identify which parts of which systems need to be replaced or upgraded. (Bredin, M. S.,2017).

8.4.1 Regular maintenance of IPS

From the study of breakdown frequency of EEPs and old pump, it has observed that the average breakdown frequency for new EEPs for last 5 years came out to be 0.23/year. (Source: Repair data for pilot project) While, old pump broken down 0.26 times per year, from survey of 18 farmers/beneficiaries. We can see that the difference between breakdown frequency of both types

of pumps is very low. Free repair can be levied under the scheme, as it puts financial burden on the scheme. Instead regular maintenance to reduce the breakdown cost can serve the purpose.

Wear rings are part of centrifugal pumps which reduces the clearance between impeller and casing. These rings are susceptible to wear with prolong usage, increasing the clearance causing leakage from high pressure side of pump. Similar thing goes for mechanical seal between shaft and pump.

8.4.2 About M&V:

Few errors have been observed in M&V process. An involvement of non-profit organization like premium technical institutions apart from hired consultants, during project monitoring and implementation can help generate unbiased, accurate, verified results.

Energy meters should be used to calculate energy savings, instead of relying on hours of pumping. These meters are need to maintained regularly.

Replacing all pumpsets from a feeder will be help us keep track of energy consumption variation from feeder level.

An attempt to collect energy consumption data from feeder has failed, as historic records were misplaced and readings seemed ambiguous. Hence, regular note of feeder readings over the project duration will be useful.

Care should be taken that measurement and calculation methodology for data collection in M&V is accurate. The details of methodology can be referred from report.

8.4.3 Miscellaneous:

For new or renewal of connections by farmers, purchase of EEPS or standardized pumpsets can be made mandatory.

A standardized pump testing setup should be made available in project locality, so the accurate results for pumpsets testing can be obtained.

Low voltage is a severe issue affecting the pump performance, utilities have to take this into account. Few, pump manufacturers design a pump motor for low voltages, but this may cause failure if voltage are improved and achieve rated desirable value.

Electricity is billed by flat rate tariff giving no direct incentive of energy savings to farmers. Many present beneficiaries had to be convince a lot to replace the pumpsets and still does not find it advantageous. So an incremental incentive through savings and free repair, instead of capital investment can be a cost effective way to deal with the situation.

Awareness campaign carried out at the beginning of such scheme, need to have its own assessment level. After getting the positive results, then only project should be taken ahead.

Chapter 9 RESULTS / CONCLUSION

9.1 Scheme Impact

9.1.1 Farmer's Response:

Primary objective of this pump replacement scheme was to save the energy with EEPS. This savings would have benefited to utilities and farmers due to reduction in electricity bills. During

implementation phase of scheme implementation agency faced many challenges one of them being convincing the farmers to participate in the scheme. But, as evident from survey farmers are reluctant to pay the bills, in addition to this the high subsidy available makes benefit of scheme unattractive to farmers. As we have seen from priority ranking the ‘service after pump purchase’ has been ranked lowest, which was said tangible benefit for scheme beneficiaries. Hence it was necessary to check with beneficiaries’ opinion of scheme. Through survey I tried to extract this information to some extent.

Figure 9.1 represents the benefits of scheme felt by the beneficiaries. They have been asked what benefit they saw through scheme, surprisingly most of them had similar opinion. It has been observed that about 80% beneficiaries from surveyed sample liked the scheme because Pumpsets are low voltage compatible. Apart from this Improved discharge compared to old pump and care free operation i.e., they never had any problem with pumpset; are other benefits felt by the farmers. While 30% of them responded to all three benefits.

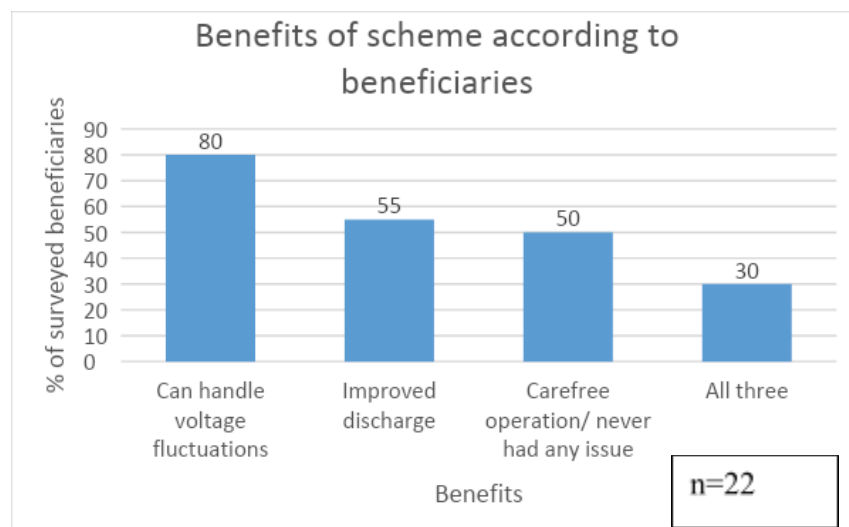


Figure 9.1 Beneficiaries’ response towards benefit of scheme

This pump replacement scheme plays a vital role in promoting the usage of energy efficient pumps. As we have seen, many beneficiaries find this scheme advantageous in some way or another. But the objective won’t be achieved if beneficiaries return back to their scenario before

the scheme. It was found that no respondents were aware of BEE star rating or energy efficient IPS, so it is less likely that they will go for star rated pumps. To verify this, they have been asked if they are going to continue the use of similar IPS of CRI brand or with IS standards, in future.

Following figure 9.2 shows, the responses obtained from surveyed beneficiaries. We can see that less than 50% of respondents replied positively. While for 5 out of 22 respondents the response was not sure or it has not been recorded. Not realising the benefit of energy saving to farmers is one of the probable reason for such response to scheme.

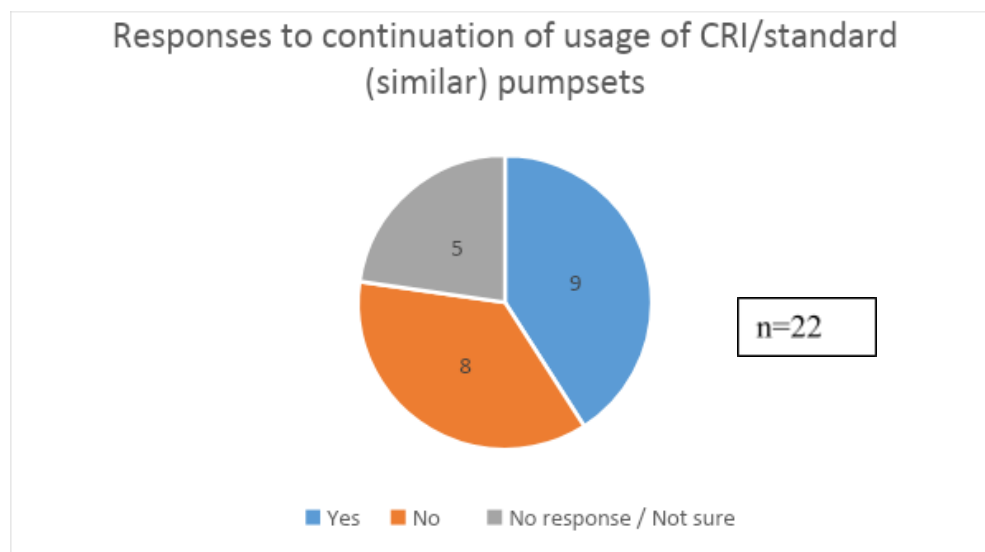


Figure 9.2 Assessing awareness about standardized pumpset among beneficiaries

Concluding Remark: There is possibility that expected outcome of scheme i.e., influencing farmers for energy savings has not been achieved. As, surveyed beneficiaries are found to be unaware about Standardized pumpsets and mostly not going to continue the use of EEPs. And they still do not care about the energy savings due to flat rate tariff, giving them no incentives.

9.1.2 Technical:

Based on primary and secondary data, following inferences has been drawn:

- At the end of 5 years of usage the average efficiency of EEPS in field obtained is 32.35% with sample size (n) of 27 and standard deviation of 13.6.
- The average voltage found out through primary data (surveys) of 27 beneficiaries is 333 V (line voltage). This is far lower than operating range, as pump are designed to operate optimally above 350V. There were only 4 points where voltage was above 380Volts and 13 with more than 350 volts. It suggests the voltage scenario in the region is very poor. And could have adversely affected the efficiency of IPS.
- Selecting appropriate capacity pumpset was one of the mandate of the pump replacement scheme. As a result, 33% (729) of EEPS were replaced with reduced HP while 2%(47) with increased HP rating. New HP rating was 10852.5 HP while old one was 11988 HP. As pumps HP was reduced by about 9.4%, it can be said that this reduced HP will definitely produce the energy savings as long as hours of pumps are not increased.
- EEPS has improved discharge, ample water is not available for irrigation and those few of farmers has already been using it for complete 8 hrs. And as seen in investigation of hours of pumping, these are very less likely to increase. Hence, we can conclude that from reduced HP rating at least 9.4% of savings are achieved.
- Before replacing the pumpsets, 826 of beneficiaries had sanctioned power HP less than what they were using. While after replacements 655 of beneficiaries has sanctioned power less than EEPS installed. As pumpsets were supposedly selected optimally, it would have been better if sanctioned power has changed accordingly. This could have helped utilities in managing appropriate load over the feeders.
- Energy savings on 5th year is about 14.48% of base, which is quite lower than targeted which was 25% approximately.
- The average power factor from measurements of M&V came to be 0.94 from secondary data. While from primary data it is found to be 0.9, but majority of farmers has removed the capacitors. M&V took the measurements by temporarily attaching the capacitor,

which may have been the reason for high power factor. While collecting primary data no capacitor was used.

- Results shown in table 9.1 are obtained based on primary and secondary data combined.

- Baseline data:

Old pumpset power consumption (kW)	6.63
Annual hours of pumping (Hr)	1640
Baseline energy consumption (kWhr)*10 ⁶	24.019
Number of pumpsets	2209

Table 9.1 Achieved energy savings

	1st M&V	2nd M&V	3rd M&V	4th M&V	5th year Primary data
Number of pumpsets sampled	308	263	252	282	27
EEPS power consumption (kW)	4.92	4.94	5.07	4.89	5.67
Total approx. energy consumption by all EEPS(kWhr)	17.82	17.90	18.37	17.72	20.54
Energy Saving in million units	6.19	6.12	5.65	6.30	3.48
Energy savings in %	25.79	25.49	23.53	26.24	14.48

Table 9.3 shows the savings in energy for feeder under the scheme. Drought may also have affected the energy consumption on these feeders.

Feeder	Year	Feeder Energy consumption MU	Comment
Bhose	2006-2009	3.44	Base - 3yr average
	2011	3.03	12% saving
	2016	2.89	16% saving
Borale	2006-2009	6.27	Base - 3yr average
	2014	4.55	27% saving
	2015	4.58	27% saving
Brahmapuri	2006-2009	6.81	Base - 3yr average
	2011	3.92	42% saving

Figure 9.3 Energy savings - feeder analysis

Concluding Remark: Voltage level are very poor which can be a cause of inefficiency in region. Actual efficiency of IPS at the end of 5 years is less than targeted but is still better than old replaced IPS. Total energy savings of 28 MU is achieved.

9.1.3 Financial:

Table 9.2 major financial tractions of the pilot project. Though it seems MSEDCL a facilitating agency for the project is in net benefit, it should be considered that the discount rate and inflation over the period is not considered. Also, there is only benefit of 3.64% to MSEDCL, the whereas average inflation rate during the project duration was more than 5%. Appendix 3 can be referred for detailed calculations.

In addition to this, if hours of pumping are different than what has considered, the benefit ratio will show a significant impact.

Table 9.2 Analysis of financial benefit of project to utility

Year	2013	2014	2015	2016	2017	Total Saving
Target Energy savings in %	38	37	36	35	34.5	
Actual Estimated energy Saving in %	25.7	25.47	23.55	26.24	14.48	
Saving (in Rs. Lakh)	167.13	165.24	152.82	174.42	93.96	

Total Payment to Agency in lakhs	109.90	110.40	103.65	118.57	65.27	507.79
MSEDCL Income (In Rs. Lakhs)	116.99	115.67	106.97	122.09	65.77	527.50
Net Profit to MSEDCL Rs. Lakh	7.09	5.27	3.33	3.52	0.50	19.21

(Source: MSEDCL for year 2013-16, Modified with 5th year data by empirical study.)

(*Not to be shared without permission of respective stakeholders)

Table 9.3 shows cost benefit analysis for implementing agency i.e., CRI pumps. From data, IA seems to be in loss of about 30.4% in terms of monetary benefit from pilot project. But there is one more perspective that should be taken into account, that is non-tangible benefit of this scheme to IA. As manufacturer had made a sell of about 2209 pumpsets through this project. In addition to this, brand awareness can be increased in the respective region, which can lead to long term economical advantage to manufacturer.

Table 9.3 Analysis of financial benefit of project to Implementation Agency

No. of pumps repaired and maintained	2568
Total repair and maintenance cost in Rs. Lakh	26.66
Approx. Cost incurred from old pump disposal Rs. Lakh	95.00
Total investment by CRI	839.42
Total revenue from MSEDCL over 5 years	507.79
Total revenue	602.79
Total expenditure	866.08
Loss in Rs. Lakhs	263.29
% loss	30.40

In Table 9.4, case 1 shows the target of project from DPR, Case 2 shows the actual achievement with this study. While case 3 shows the financial feasibility of scheme with corrected methodology of calculation and parameters like hours of pumping. It is observed planned properly scheme is financially viable. (Cost of energy (4.4) = Average cost of energy (6.5)- cost of energy for farmers (2.1) in Maharashtra) ((MERC,2016) (MSEDCL tariff booklet)

Few reasons targeted financial savings are not achieved

- Initial investment risen almost by 100%.
- Financial feasibility was calculated at Rs. 3.65/unit. (DPR, 2009)
- Measured efficiencies are lower than estimated by manufacturer.

Table 9.4 Financial feasibility analysis

Assumed Base EEPS eff. 39.68%	Case 1 - Targeted	Case 2 - Actual	Case 3 - With corrected parameters
Hours of pumping for baseline	1640	1640	1200
Hours of pumping for for EEPS (base)	1640	1640	980
Cost of energy	3.65	2.7	4.4
Efficiency drop rate (%/year)	-	-	1.45
Method used for calculating energy consumption	Power rating of pump		Efficiency
Number of years savings considered	5+	5	10
Efficiency at the end of years in %	34.50%	34.50%	25.18
Estimation of Energy Savings MU	44.2	27.73	39.5
Financial gain %	28%	-13%	73%
IRR	10%	-6%	18%

Concluding Remark (financial):

In general, the total expenditure was about 866 lakhs, while total monetary benefit (without considering inflation etc.) was about 848 lakhs i.e., the loss of approximately 2%. While assuming 10% discount rate and 5% inflation each year, the loss is 13% with IRR of -6% approximately (Details can be found in annexure 3). Considering 1200 hours of pumping project would have been in 33% loss. With corrected methodology project is observed to be beneficial.

9.2 Conclusion:

Pump replacement project has been theoretically effective in saving the energy. And it is definite that EEPS is an effective tool to save energy for longer duration. The financials of project shows the loss although the non-tangible benefit like branding of manufacturer may prove to be important aspect in long term gain for manufacturer. Efficiency of EEPS in field after 5 years is found to be 32.34% which is less than M&V targeted value of 34.5%. Although the baseline efficiency was 22% hence energy savings are going to continue for some more duration resulting gain for utilities.

The quality of power supply is a critical issue that need to be addressed quickly, as it not only affects the pump selection by farmer but also is a cause for inefficiency of IPS. Low voltage compatibility and durability are given utmost importance while pump selection by farmers.

There is still no awareness about BEE star rating for pumps among scheme beneficiaries. Hours of pumping has not been reduced and bills are at default. All this proves the indifference among farmers towards energy efficiency. Hence increasing the awareness among farmer is need of a time. Electrical connection with only EEPS/ star rating can be made mandatory. Low voltage compatible pumps can be used to deal with issue of lower voltage in fields.

Performance of pumpsets deteriorated with time and poor voltage. Hence consideration of these aspect for planning similar project will play a crucial role in scheme financials. Annual hours of pumping considered for project (i.e., 1640) found to be an overestimate. And as power rating and hours is considered for energy saving calculation, the results obtained are questionable. Usage of efficiency for calculating energy savings takes care of change in hours of pumping as well as variation in input electrical power. Analysis with corrected methodology, project can seem financially viable.

M&V was done by third party but observed errors through this study raises questions on results obtained. The raw data of this pilot project revealed these serious errors which raises question on results obtained by previous AgDSM projects as well. Hence, monitoring and verification process should be made more stringent. Voluntary Involvement of technical institution or non-profit organizations can help in achieving accurate, unbiased and verified results. Continuation and expansion of project with ESCOs can help in achieving the energy saving. Though modifications are needed during implementation to make scheme more successful and financially feasible.

9.3 Future scope:

- Precise study of effect of voltage and time on Performance of Pumpsets.
- Economic analysis for various combinations of rectification measure such as, addition of capacitor, replacement with EEPS, free repair and maintenance in existing pumps, managing feeder load etc.
- Detail study for improving the BIS standard of efficiency of IPS.

- Similar scheme/pump replacement project analysis at different geographic location where the scheme has been implemented. To see if there are any variation regarding scheme implementation and assess the need of customization accordingly.
- Preparing a protocol to deduce the hours of pumping in irrigation.
- Investigating the quality of voltage in field – Variation, reasons, possible improvements

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APPENDIX 1

Calculation Methodology:

Electric input, $P_T = (V_1 * I_1 * P.F. + V_2 * I_2 * P.F. + V_3 * I_3 * P.F.) / (\sqrt{3})$

Where,

P_T = Total Power (W)

V_n = Line voltage (V)

I_n = Current (A)

P.F. = power factor

Hydraulic Output, $P_H = \rho * Q * g * H_T$

Where,

ρ = density of fluid in (kg/m³)

Q = flow rate (m³/s)

g = 9.81 m/s

H_T = Total dynamic head in (m)

$H_T = \text{Static head } (h_s) + \text{frictional head } (h_f)$

Static head = Total elevation between water surface and discharge tapping point.

Frictional head h_f = friction (major) loss in pipe + friction (minor) loss at valves and fittings

$$\text{Major losses} = \sum \frac{fLv^2}{2gD}$$

$$\text{Minor losses} = \sum \frac{k_i v^2}{2g}$$

Where,

f = friction factor

k_i = minor loss coefficient

L = Length of pipe (m)

D = Diameter of pipe (m)

V = Velocity (m/s)

Finding friction factor f (turbulent flow)

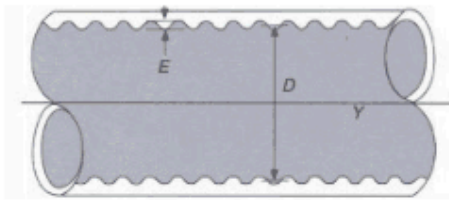
$$1. V = \frac{Q}{A} = \frac{4Q}{\pi D^2}$$

$$2. \text{ Reynold's number (Re)} = \frac{\rho V D}{\mu} \quad ; \quad \mu = \text{Dynamic viscosity in kg/m.s}$$

3. Search for value of relative roughness (ϵ) for particular material of pipe. And find the ratio of (ϵ/D)

4. Use Moody's chart to find f for value of (Re) and (ϵ/D).

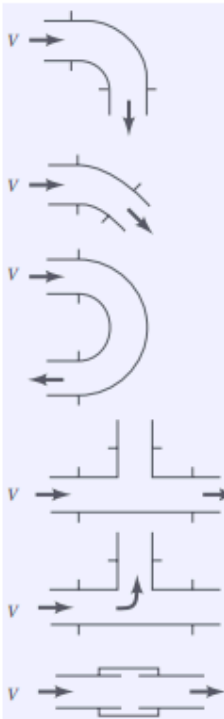
OR [Can directly use this link <http://www.freecalc.com/fricfram.htm>]



Material	Absolute Roughness (mm)
Copper, Lead, Brass, Aluminum (new)	0.001 - 0.002
PVC and Plastic Pipes	0.0015 - 0.007
Flexible Rubber Tubing - Smooth	0.006-0.07
Stainless Steel	0.0015
Steel Commercial Pipe	0.045 - 0.09
Weld Steel	0.045
Carbon Steel (New)	0.02-0.05
Carbon Steel (Slightly Corroded)	0.05-0.15
Carbon Steel (Moderately Corroded)	0.15-1
Carbon Steel (Badly Corroded)	1-3
Asphalted Cast Iron	0.1-1
New Cast Iron	0.25 - 0.8
Worn Cast Iron	0.8 - 1.5
Rusty Cast Iron	1.5 - 2.5
Galvanized Iron	0.025-0.15
Wood Stave	0.18-0.91
Wood Stave, used	0.25-1
Smoothed Cement	0.3
Ordinary Concrete	0.3 - 1
Concrete – Rough, Form Marks	0.8-3

(Source: Nuclear-power.net)

Finding minor loss coefficients K_i

a. Elbows		
Regular 90°, flanged	0.3	
Regular 90°, threaded	1.5	
Long radius 90°, flanged	0.2	
Long radius 90°, threaded	0.7	
Long radius 45°, flanged	0.2	
Regular 45°, threaded	0.4	
b. 180° return bends		
180° return bend, flanged	0.2	
180° return bend, threaded	1.5	
c. Tees		
Line flow, flanged	0.2	
Line flow, threaded	0.9	
Branch flow, flanged	1.0	
Branch flow, threaded	2.0	
d. Union, threaded		0.08
*e. Valves		
Globe, fully open	10	
Angle, fully open	2	
Gate, fully open	0.15	
Gate, $\frac{1}{4}$ closed	0.26	
Gate, $\frac{1}{2}$ closed	2.1	
Gate, $\frac{3}{4}$ closed	17	
Swing check, forward flow	2	
Swing check, backward flow	∞	
Ball valve, fully open	0.05	
Ball valve, $\frac{1}{2}$ closed	5.5	
Ball valve, $\frac{3}{4}$ closed	210	

*See Fig. 8.36 for typical valve geometry

(Source: <https://vaoengineering.wordpress.com/2012/12/30/head-loss-coefficients/>)

Following are the assumption made for the calculations of minor losses:

Relative roughness	PVC & HDPE	5E-06
	GI E(m)	0.0008
Assumed K values	elbow joints	0.3
	T- joint	0.2
	valves	2
	chamber	1

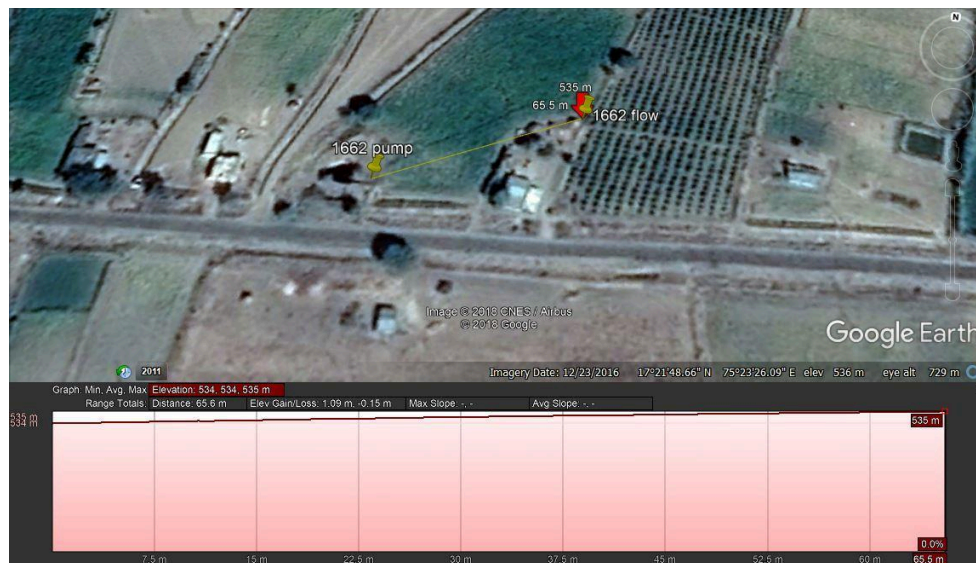
Measurements' methodology:

Head Measurements:

Depth of water surface is measured with the Plumb bob.

For friction head pipe length is required, hence distance between pump and discharge/ tapping point is measured with GPS and google earth software. Same was used to measure the elevation. The co-ordinated at locations were recorded and inputted into google earth, based on visual observations of field coordinates were adjusted to reduce the error.

Following image shows an example of measurement.



Discharge:

A water container and stopwatch method is used to measure the discharge.



Electrical Measurements:

“Chauvin Arnoux Krykard F09” Clampmeter & “Nanovip Elcontrol VIP Energy Power Meter” are used for the electrical measurements.

APPENDIX 2

Findings from M&V data:

Increased efficiency suggests lower power consumption for same output power or/& hp rating. During few calculations we needed the difference between power ration of local pumpsets and EEPSS. Through surveys done under the scheme we had this data available. Hence few generic findings from it are mentioned in **TABLE NO.** It shows the power consumption by IPS with different hp ratings. From the study of different pump catalogues it has found that generally pumps are designed to operate optimally at voltage as low as 350 V. Hence, following sample are the measurements taken at voltage 350V during M&V. With load the input power consumption of IPS varies hence the results obtained are approximations and not accurate. Also, the standard deviation is low, data is normally distributed and outliers are omitted which has improved the precision of data.

Old/Local IPS				
Readings at	Pump Rating	Sample No.	Average kW	Standard Deviation
Voltage >350	3hp	44	3.75	0.52
	5hp	150	6.34	0.99
	6hp	30	7.59	0.54
Voltage >380	7.5hp	31	9.12	1.28
	10hp	7	10.20	1.05
EEPS				
Voltage >350	3hp	63	3.10	0.35
	5hp	158	5.07	0.62
	6hp	37	6.24	0.47
Voltage >380	7.5hp	27	7.60	0.76
	10hp	5	8.80	1.39

APPENDIX 3

1. Details of error in M&V measurements.

Sr . N o.	CRI ID	Set A - M&V (Secondary data)			Set B - Our's/ corrected (Primary data)			% error (100* (B-A) /B)	Set C -	%	Pump type
		Len gth (m)	Slo pe (m)	Efficie ncy %	Len gth (m)	Slo pe (m)	Efficie ncy %		With	error	
									metho d	(100* (B-C) /B)	
1	1806	670	0	60.98	153	3	26.15	-133	30.76	-17.6 3	Openw ell

2	1016	396	1.2 2	73.16	91	3	45.93	-59	47.85	-4.18	Openwell
3	340	300	0	48.74	168	1	33.83	-44	25.57	24.42	Monoblock
4	743	326	2.7	42.18	71	1.6	21.48	-96	26.39	-22.8 6	Openwell
5	1731	447	2.7	28.96	140	-0. 5	20.93	-38	12.83	38.70	Monoblock
6	2326	58	0.6	40.29	40	1.5	38.99	-3	27.16	30.34	Openwell
7	1808	412	1.2	25.04	181	2.6	24.83	-1	22.53	9.26	Borewell
8	742	386	2	16.28	412	9	23.15	30	13.57	41.38	Monoblock
9	152	406	3	16.66	406	6.5	19.37	14	15.59	19.51	Openwell
	Mean			39.14			28.30	-36.8 7	24.69	13.22	
	Std.Dev.			18.46			8.69		10.19		

2. Sensitivity Analysis Details:

I/P power (kW)	Q (lps)	static head = water depth+ elevation	Dia. of pipe 1 (m)	Length of pipe 1 (m)	Dia. Of pipe 2 (m)	Length of pipe 2 (m)	Hf for pipe 1 (m)	Hf for pipe 2 (m)	O/P power (kW)	Efficiency %	% incremental error in Eff.
Example 1: CRI ID 1016; Pump Model- CSM2; 5HP openwell											
4.74	6.99	26.00	0.06	30.48	0.06	61.00	2.06	4.13	2.21	46.57	

4.74	6.99	26.00	0.06	30.48	0.06	162.0 0	2.06	10.96	2.68	56.45	21.23
4.74	6.99	26.00	0.06	30.48	0.06	263.0 0	2.06	17.80	3.14	66.34	17.51
4.74	6.99	26.00	0.06	30.48	0.06	364.0 0	2.06	24.63	3.61	76.23	14.90
Example 2: CRI ID 742; Pump Model- H41-1; 5HP openwell											
4.72	4.26	21.50	0.06	15.00	0.08	392.0 0	0.41	4.49	1.10	23.37	
4.72	4.26	21.50	0.06	15.00	0.08	292.0 0	0.41	3.34	1.06	22.36	-4.33
4.72	4.26	21.50	0.06	15.00	0.08	192.0 0	0.41	2.20	1.01	21.35	-4.53
4.72	4.26	21.50	0.06	15.00	0.08	92.00	0.41	1.05	0.96	20.33	-4.75

3. Analysis of financial benefit of project to utility (table 9.2)

Year	2013	2014	2015	2016	2017	Total Saving
Target savings in %	38	37	36	35	34.5	
Actual Saving in %	25.7	25.47	23.55	26.24	14.48	
Deficit in %	32.37	31.16	34.58	25.03	58.03	
Penalty in Rs lakhs	28.60	27.53	30.56	22.12	51.27	
(Maintenance (Rs. 88lakh/yr)- penalty) in lakh	59.76	60.83	57.80	66.24	37.09	
Saving in Million units	6.19	6.12	5.66	6.46	3.48	
Saving (in Rs. Lakh)	167.13	165.24	152.82	174.42	93.96	
Payment to CRI (30% Share)	50.139	49.572	45.846	52.326	28.188	
Total Payment to Agency in lakhs	109.90	110.40	103.65	118.57	65.27	507.79
MSEDCL Income (In Rs. Lakhs)	116.99	115.67	106.97	122.09	65.77	527.50
Net Profit to MSEDCL Rs. Lakh	7.09	5.27	3.33	3.52	0.50	19.21

Imported from MSEDCL for year 2013-16) Modified with 5th year data by empirical study.

(*Not to be shared without permission of respective stakeholders and author)

4. Financial Viability with NPV calculation:

	Year 1	Year 2	Year 3	Year 4	Year 5
Discount rate d	0.1				
Inflation rate	0.05				
x	0.95	0.91	0.87	0.83	0.79
electricity Per unit charges Rs.	2.7				

old pump power rating	6.63				
Baseline energy consumption in MU	24				
No. of pumps	2209				
Annual hours of pumping	1640				
Average power rating of each pump (kW)	4.92	4.94	5.07	4.89	5.67
Energy consumption by new pumps (MU)	17.82	17.89	18.36	17.71	20.51
Total energy savings (MU)	6.19	6.12	5.65	6.3	3.47
Revenue from old pump Rs. Lakh	95				
Total revenue generated per year through energy savings Rs. Lakh	167	165	153	170	94
NPV of revenue Rs. Lakh	159.65	150.6	132.7	141.2	74.4
Total NPV of Revenue Rs. Lakh	749				
Initial Investment Rs. Lakh	839.4				
Approx. Annual repair expense Rs. Lakh	5.33				
Total NPV of repair cost Rs. Lakh	23				
Total NPV Investment Rs. Lakh	863				
loss %	13				
IRR	-0.06				

APPENDIX 4

Survey Questionnaire :

Name of farmer:.....

Date:.....

CRI ID:.....Sanctioned load:.....Meter status:.....

Village:.....Feeder:.....Co-ordinates:.....

Pump rating: (Old:.....; New:.....) Date of purchase:.....Type:.....

Total land holding:.....Irrigated land:.....Flood/drip/other

Technical Measurements:

	R	Y	B
Voltage			
Current			
Power factor			
Power			

Flow rate		
Depth of water level from surface		
Depth of pump from surface		
Length of pipe		
Diameter of pipe		
Material of pipe		
Other details		

Pump condition check: (take photos)

Pump	Good/ bad/ Y/N	Panel	Good/ bad	Supply/ system/ piping leakage	Good/ bad
Star rating		Meter working status			
Scaling		Fuse			
Do they maintain		Wiring			
		Capacitor (bypass phase)			

No. of breakdowns from date of purchase..... Types of breakdown.....

No. of breakdown of old pumps per year.....

What are the pump breakdown reasons according to you?

.....
.....

No. of hours of pumping over the year (current).....

Have hours of pumping reduced? **(Yes/No)** What was hours of pumping before?.....

How much electricity bill you pay now vs used to pay for old pump?

.....
.....

How do you select the pump?.....

Priorities while selecting the pump						
Cost	Low Voltage compatibility	Quality/ standards/ Brand	Guarantee	After sales service	Durability	Other

Which pump is better old one or new pump(EEPS)? Why? Name and rating of old pump.

.....
.....
.....
.....

Do you know about star rating system? If Yes, what do know about it?

.....
.....

Will you continue to use or buy star rated (branded) pump? Why?

.....
.....
.....
.....

What difference did you find between old and new (EEPS) pump?

.....
.....

Benefits of scheme according to you:

.END

M&V Survey Form – (sample survey)

IS/ISO 9001:2008 Certified
MITCON
CONSULTANCY & ENGINEERING
SERVICES LTD.

Data Collection Format - Fifth Year M & V Report 2017-18

M & V Site Data Collection for AGDSM Project, Mangalwedha - Solapur.

Name of the Farmer :-		Date:	8-2-18
CRI ID. Number	1016	Village:	
MSEDCL Consumer No.		Feeder:	
Meter Sr. Number	5144145	Meter Reading	—

A: General Field Data

Pump Location	Well mounted	Borewell Pump	River Bed	Lake Cannel
Pump Type	Mono Block	Well Submersible	Borewell Submersible	

B: Post Replacement New Pump Details

Make:	CRI	Pump Rating HP	5HP
Motor I/P Power		B- Phase	Average
Voltage (V)	327	337	330
Current (A)	9.37	8.44	8.73
Input Power (KW)	5.02	4.79	4.51
Power Factor	0.95	0.95	0.98
Tank Size	158	10.5	16.49
	Its	sec	sec

Well/Bore/River /Cannel/Lake Depth	95	ft	Depth of Pump Installation from GL	95	ft	Depth Water Table from GL	70	ft
Suction 1 (material)			Suction Diameter 1		mm	Suction Length 1		ft
Suction 2 (material)			Suction Diameter 2		mm	Suction Length 2		ft
Delivery 1 (material)	HDCP		Delivery Diameter 1	2	mm	Delivery Length 1	100	ft
Delivery 2 (material)	PVC		Delivery Diameter 2	2	mm	Delivery Length 2	1200	ft
T-Joint (Equivalent)			Elbow Joint (Equivalent)					
Valves (Equivalent)			Chambers (Equivalent)					
Foot Valve Type			Pipe Slope Head From Pump Installation to End Point				04	ft

C: New Installation Status Checks

Meter Status	Remark Running/Burn/Faulty/Bypass	
Pump	Remark Running/Faulty/Rewinding	
Capacitor Bank kVAR	Remark Running/Burn/Faulty/Bypass	3KVA

D: Additional Observations

1	
2	
3	
4	

Checked by:

For CRI		For MITCON	
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Verified by:

For MSEDCL		For Farmer	धा.मि. अ. अ. दि. अ. अ. 2018 2018
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Field Measurements: Primary Data

CRI ID	Pump rating	Pump type	Pump model	Avg. Voltage	Avg. PF	Discharge in lps	Total head (m)	Friction head (m)	Efficiency %
808	3	Well-sub.	CSM-1	365	0.83	7.8	9.4	5.7	18.91
152	5	Well-sub.	CSM-2	356	0.98	6.03	24.5	9	19.37
1662	5	Well-sub.	CSM-2	421	0.83	6.15	21.5	7.4	29.93
2326	5	Well-sub.	CSM-2	380	0.91	11.61	19	6.9	40.74
1016	5	Well-sub.	CSM-2	331	0.93	6.99	31.77	6.2	45.93
743	5	Well-sub.	CSM-3S(5)	318	0.94	6.93	18.53	4.7	21.48
1637	5	Well-sub.	CSM-3S(5)	246	0.9	6.5	20.39	14.4	23.53
1806	5	Well-sub.	CSM-3S(5)	370	0.98	7.14	24.4	10.7	25.29
709	5	Well-sub.	SR65 4HP/8	372	0.89	6.67	24.15	5	35.26
341	5	Monoblock	5H3	352	0.9	8.5	27.11	17	33.83
1731	5	Monoblock	H40-1	195	0.95	5.26	17.15	5.4	20.93
99	5	Monoblock	H40-1	336	0.92	6.57	16.87	5.7	23.15
742	5	Monoblock	H40-1	366	0.94	4.26	26.09	4.9	23.15
2260	7.5	Borewell	SM60 7.5HP/8	309	0.95	6.78	50.08	12.5	41.6
831	5	Borewell	SR61 5HP/5	330	0.82	4.69	18.3	33.4	67.09
1898	3	Borewell	SR63 3HP/5ST	362	0.84	4.52	42	14.8	48.74
36	5	Borewell	SR65 5HP/10	329	0.98	3.93	69.3	13.4	53.67
1655	6	Borewell	SR65 6HP/12	319	0.95	2.09	79.03	4.9	24.94
2160	3	Borewell	SR66 3HP/8	409	0.66	1.86	30.5	1.1	14.34
2149	6	Borewell	SR67 6HP/20	365	0.99	2.88	129.5	6.6	42.73
1808	7.5	Borewell	SR67 7.5HP/25	358	0.91	2.31	80.43	7.7	24.83
			Average	342	0.9	5.7	37.1	9.4	32.4
			SD	50	0.1	2.4	29.4	6.9	13.6